THE ASSAM
ROYAL GLOBAL UNIVERSITY
GUWAHATI

ROYAL SCHOOL OF ENVIRONMENTAL AND EARTH SCIENCES (RSEES)

DEPARTMENT OF GEOGRAPHY AND GEOINFORMATICS

COURSE STRUCTURE & SYLLABUS (BASED ON NATIONAL EDUCATION POLICY 2020)

FOR

B.A/ B.Sc. GEOGRAPHY
(4 YEARS SINGLE MAJOR)

W.E.F

AY - 2026– 27

B.A/B.Sc. (H) Geography
Programme Structure
YEAR- 1

Semester 1			
Sl. No	**Subject Code**	**Names of subjects**	**Credit**
		Major Course	
1	GEO162M101	Geography of Human and Cultural Landscape	3
2	GEO162M102	Geomorphology	3
		Minor	
3	GEO162N101	Fundamentals of Physical Geography	3
		Interdisciplinary	
4	IKS992K101	Indian Knowledge System-I	3
		Ability Enhancement Compulsory Course	
5	CEN982A101	Communicative English	2
	BHS982A102	Behavioral Science -I	
		Skill Enhancement Courses (SEC)	
6	GEO162S111	Basics of Cartography	3
		Value Added Courses (VAC)	
7		Basket course	3
		Swayam Course(s)	
8		Upcoming	3/4
		TOTAL	**20+3/4**
Semester - 2			
Sl. No	**Subject Code**	**Names of subjects**	**Credit**
		Major Courses	
1	GEO162M201	Climatology and Oceanography	3
2	GEO162M202	Geography of Tourism	3
		Minor	
3	GEO162N201	Fundamentals of Human Geography	3
		Interdisciplinary	
4	IKS992K201	Indian Knowledge System -II	3
		Ability Enhancement Compulsory Course	
5	CEN982A201	Communicative English	1
	BHS982A202	Behavioral Science -II	1
		Skill Enhancement Courses (SEC)	
6	GEO162S211	Fundamentals of Geoinformatics	3
		Value Added Courses (VAC)	
7	VAC992V2015	Geography of India	3
		Swayam Course(s)	
8		Upcoming	3/4
		Total	**20+3/4**

YEAR- 2

	Semester -3		
Sl. No	**Subject Code**	**Names of subjects**	**Credit**
		Major Course	
1	GEO162M301	Economic Geography	4
2	GEO162M302	Soil and Biogeography	4
		Minor	
3	GEO162N301	Biogeography	4
		Interdisciplinary	
4	GEO162I301	Agricultural Practices in India	3
		Ability Enhancement Compulsory Course	
5	CEN982A301	Communicative English -III	1
	BHS982A302	Behavioral Science -III	1
		Skill Enhancement Course	
6	GEO162S3111	Introduction to Quantitative Techniques	3
		Swayam Course(s)	
7		Upcoming	3/4
		TOTAL	**20+3/4**
	Semester 4		
Sl. No	**Subject Code**	**Names of subjects**	**Credit**
		Major Courses	
1	GEO162M401	Social and Political Geography	4
2	GEO162M402	Environmental Geography	4
3	GEO162M403	Indian Knowledge System on Geography (IKS)	4
		Minor	
4	GEO162N401	Economic Geography	3
5	GEO162N402	Regional development	3
		Ability Enhancement Compulsory Course	
6	CEN982A401	Communicative English -III	1
	BHS982A402	Behavioral Science -III	1
		Swayam Course(s)	
7		Upcoming	3/4
		TOTAL	**20+3/4**

YEAR- 3

	Semester 5		
Sl. No	**Subject Code**	**Names of subjects**	**Credit**
		Major Course	
1	GEO162M501	Regional Planning and Development	4
2	GEO162M502	Population and Settlement Geography	4
3	GEO162M503	Disaster Management	4
		Minor	
4	GEO162N501	Population and Political Geography	4
		Internship	
5	GEO162M524	4 weeks or 120 hours	4
		TOTAL	**20**

Sl. No	**Subject Code**	**Names of subjects**	**Credit**
		Major Courses	
1	GEO162M601	Geography of India	4
2	GEO162 M602	Geographical Thought	4
3	GEO162 M603	Agricultural Geography	4
4	GEO162 M614	Cartographic techniques in Geography	4
		Minor	
5	GEO162N601	Geography of Development of India	4
			20

YEAR- 4

		Semester 7	
Sl. No	**Subject Code**	**Names of subjects**	**Credit**
		Major Course	
1	GEO162 M701	Geography of Rural Development	4
2	GEO162 M702	Urban Geography	4
3	GEO162 M703	Geography of North East India and Assam	4
4	GEO162 M714	Field Techniques in Geography	4
		Minor	
5	GEO162N701	Geography of North East India	4
		TOTAL	**20**

		Semester 8	
Sl. No	**Subject Code**	**Names of subjects**	**Credit**
		Major Courses	
1	GEO162M801	Research Methodology	4
2	GEO162M822	Research I	12
3	GEO162N801	Environment and Sustainable Development	4
400 level advance core course		In lieu of Research	
	GEO162M804	Global Climate Change	4
	GEO162M805	Environment and Sustainable Development	4
	GEO162M806	Geography of Health	4

Semester 1

Semester 1
Major course

Paper I Core Course	GEOGRAPHY OF HUMAN AND CULTURAL LANDSCAPE L-T-P-C: 3-0-0-3 Scheme of Evaluation: (T) Course Level : 100	Subject Code: GEO162M101

Course Objectives: *The objective of this course is to enable students to understand the chronological development of Human Geography through the contributions of various scholars, approaches, and schools of thought, along with the major themes and components of Cultural Geography. The course also aims to promote awareness of global development concerns, particularly SDG 11: Sustainable Cities and Communities and SDG 10: Reduced Inequalities, by encouraging students to critically examine issues of urban sustainability, cultural diversity, social inclusion, and spatial inequalities within human societies.*

Course Outcomes:

After successful completion of the course, the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Define** the various parameters and components of the sub-branch.	BT1
CO2	**Interpret** the development of the humanistic view of Geography with special reference to SDG 11: Sustainable Cities and Communities through inclusive, resilient, and culturally sustainable human environments.	BT2
CO3	**Identify** the various aspects of human geography.	BT3
CO4	**Discover** the humanistic perspective and its dimensions in Geography in relation to the physical and cultural surrounding fostering the ideals of SDG 10: Reduced Inequalities through a deeper understanding of human experiences and inequalities across different communities and regions.	BT4

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Meaning, contents and Scope of Human Geography, human versus physical geography; branches of Human Geography; Development of Human Geography; Contributions of German and French Geographers.	15
Unit 2	Approaches to the study of human geography: Determinism, possibilism, human ecology, positivism, **S**chools of human geography: Ecology and landscape	15
Unit 3	Definition, Scope and Approaches of Cultural Geography; Major themes of Cultural Geography: Concept of cultural hearth, cultural region, cultural landscape and cultural integration.	15
Unit 4	Characteristics of culture, its components, and functions; Cultural diffusion and factors associated with it, Major cultural regions of the World and their characteristics.	15
	Total	**60**

Notional Credit Hours for the course: 30X3= 90

Total credit in the paper	Lecture/Tutorial	Experiential Learning
3	60 hours	30 hours
		Field visit for studying landscapes with geo-cultural significance

Text Books:

1. Huntington, E., 1951: ***Principles of Human Geography***, John Wiley & Sons, Inc, New York
2. Hussain, M., 1994: ***Human Geography***, Rawat Publication, New Delhi.

Reference books:

1. Haggett, P., 1972: Geography: A Modern Synthesis, Harper & Row, New York
2. Singh, S., 1991: Environmental Geography, PustakBhawan, Allahabad
3. Strahler, A.N. & A.H. Strahler, 1976: Geography and Man's Environment, John Willey, New York
4. Knowles, R. and Wareing, J., 1990: Economic and Social Geography, Rupa Publications India

Paper II Core Course	L-T-P-C: 3-0-0-3	GEOMORPHOLOGY Scheme of Evaluation: (T) Course Level : 100	Subject Code: GEO162M102

Course Objectives: *The pivotal point of this course is to make students familiar with the fundamental concepts of Geomorphology, incorporating topics related to geomorphic structures and processes, Earth's interior and composition, and the evolution of landforms. The course also aims to develop awareness regarding environmental sustainability, climate resilience, and ecosystem conservation in alignment with SDG 13: Climate Action and SDG 15: Life on Land through the study of Earth surface processes and their impacts on terrestrial environments.*

Course Outcomes:

After successful completion of the course, the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Define** the functioning of Earth systems in real time with an understanding of environmental changes and their implications for SDG 13: Climate Action.	BT1
CO2	**Outline** the roles of structure, stage, and time in shaping landforms along with **interpreting** geomorphological maps, while recognizing their significance in ecosystem conservation and SDG 15: Life on Land.	BT2
CO3	**Apply** geomorphological knowledge in geographical research related to environmental management, climate adaptation, and sustainable land-use practices.	BT3
CO4	**Distinguish** between the mechanisms controlling geomorphic processes and **analyses** how natural and anthropogenic factors influence landform development, environmental degradation, biodiversity, and climate-related challenges.	BT4

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Geomorphology: Nature, Scope, key concepts and theories of landform development, Systems approach	15
Unit 2	Composition of the Earth with special reference to seismology; Earth's Interior and its Structure, Earth Movements: Isostasy, Plate Tectonics, Types of Folds and Faults, Earthquakes and Volcanoes.	15
Unit 3	Geomorphic Processes: Weathering, Mass Wasting, Cycle of Erosion (Views of Davis and Penck); Profile drawing (Serial, Superimposed, Composite, Projected) -	15
Unit 4	Evolution of Landforms (Erosional and Depositional): Fluvial, Karst, Aeolian and Glacial; Applied Geomorphology; Drainage basin delimitation, Ordering of streams, calculation of bifurcation ratio, length ratio, computation of basin circularity ratio	15
	Total	**60**

Notional Credit Hours for the course: 30X3= 90

Total credit in the paper	Lecture/Tutorial	Experiential Learning
3	60 hours	30 hours
		Field visit for studying landforms of geomorphic significance

Text Books:

1. Ahmed, E., 1985: ***Geomorphology***, Kalyani Publishers, New Delhi
2. Singh, Savindra., 1998: ***Geomorphology***, Pravalika Publications, Allahabad, Uttar Pradesh.
3. Thornbury W.D., 2004: Principles of Geomorphology, CBS, India

Reference Books:

1. Bridges E. M., 1990: *World Geomorphology*, Cambridge University Press, Cambridge.
2. Dayal, P. (2nd Ed.) 1996, A Textbook of Geomorphology, Shukla Book Depot, Patna
3. Khullar D.R. 2012: Physical Geography, Kalyani Publishers, New Delhi
4. Bloom A. L., 2003: ***Geomorphology: A Systematic Analysis of Late Cenozoic Landforms***, Prentice-Hall of India, New Delhi.

Minor

Paper: Minor course	FUNDAMENTALS OF PHYSICAL GEOGRAPHY L-T-P-C: 3-0-0-3 Scheme of Evaluation: (T) Course Level : 100	Subject Code: GEO162N101

Course Objectives: *The course aims to make students aware of physical surroundings, including landforms, climate, ecosystems, and oceanic landforms, along with their processes and patterns on the Earth's surface. It also seeks to develop knowledge on ecological balance, global climatic changes, environmental sustainability, and their consequences in alignment with SDG 13: Climate Action and SDG 15: Life on Land through the understanding of climate systems and terrestrial ecosystem conservation.*

Course Outcomes:

After successful completion of the course, the students will be able to:

CO	Course Outcome	Blooms Taxonomy Level
CO1	**Define** the basic terms and terminologies related to the physical Earth with reference to environmental sustainability and climate systems.	BT1
CO2	**Compare** different global climatic patterns, climate change, and their related consequences while understanding their relevance to SDG 13: Climate Action.	BT2
CO3	**Identify** physical processes and the resultant environments that shape life on planet Earth, with emphasis on ecosystem conservation and SDG 15: Life on Land.	BT3
CO4	**Examine** ecological, climatic, and atmospheric phenomena of the Earth and analyse their impacts on biodiversity, land ecosystems, and climate resilience.	BT4

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Nature, Scope and branches of Physical Geography; Processes of landform development - Exogenic and endogenic processes; Earth materials- Composition of the earth's crust, Rocks and minerals (3); Concept of Cycle of Erosion (1)	15
Unit 2	The lithosphere and Plate Tectonics; Distribution of plants and animals, Structure, functioning and material cycles of Ecosystem, Ecological Balance, Traditional ecological knowledge	15
Unit 3	Elements and factors of weather and climate; Structure and Composition of Atmosphere; Air Circulation, Pressure Systems, Cyclones and anticyclones, Global Climatic patterns and Climatic change and its consequences.	15
Unit 4	Bottom Configuration of oceans with special reference to the Atlantic Ocean, Distribution of salinity, temperature and ocean deposits and resources, and ocean Currents.	15
		60

Notional Credit Hours for the course: 30X3= 90

Total credit in the paper	Lecture/Tutorial	Experiential Learning
3	60 hours	30 hours
		Field work and report making

Text Books:

1. Ahmed, E., 1985: ***Geomorphology***, Kalyani Publishers, New Delhi
2. Bloom A. L., 2003: ***Geomorphology: A Systematic Analysis of Late Cenozoic Landforms***, Prentice-Hall of India, New Delhi.
3. Singh, Savindra., 1998: ***Geomorphology***, Pravalika Publications, Allahabad, Uttar Pradesh.

Reference Books:

1. Bridges E. M., 1990: *World Geomorphology*, Cambridge University Press, Cambridge.
1. Dayal, P. (2nd Ed.) 1996, A Textbook of Geomorphology, Shukla Book Depot, Patna
2. Khullar D.R. 2012: Physical Geography, Kalyani Publishers, New Delhi

Semester – I		
Paper I/Subject Name: Introduction to Indian Knowledge System - I		
Subject Code: IKS992K101		**L-T-P-C – 2-1-0-3**
Credit Units: 3		**Course Level: 100**
Scheme of Evaluation: Summative Assessment - 70% + Formative Assessment -30%		
Credit Distribution (NCH)		
Lecture/Tutorial (Hours)	**Practical (Hours)**	**Experiential Learning (Hours)**
60	0	30

Course objectives:

This Foundation course is designed to present an overall introduction to all the streams of IKS relevant to the UG program. It would enable students to explore the most fundamental ideas that have shaped Indian Knowledge Traditions over the centuries.

Course Outcomes:

On completion of this course students will be able to :

CO	Contents	BT Level
CO_1	Recall the rich heritage of Indian knowledge systems	BT level 1
CO_2	Describe the contribution of Indian knowledge systems to the world	BT level 2
CO_3	Demonstrate knowledge of sociocultural and ethnolinguistic diversity that constitutes the soul of Bharatvarsha	BT level 2
CO_4	Apply traditional knowledge and techniques in day-to-day life	BT level 3
CO_5	Distinguish knowledge traditions that originated in the Indian subcontinent	BT level 3

Module	Course Contents	Periods
I	**Introduction to Indian Knowledge Systems (IKS):** About Indian Knowledge System; Definition of Indigenous/ Traditional Knowledge; Scope, and Importance of Traditional Knowledge. **Ancient India- Bharat Varsha:** People of Ancient Bharat Varsha; Our great natural heritage: The great Himalayas and the rivers; The civilizations of the Sindhu-Ganga valley, and the Brahmaputra valley; Our coastal plains; Our Nature: Forests and Minerals; Ancient Indian Traditional Knowledge and Wisdom about nature and climate.	15
Module	**Course Contents**	**Periods**
II	**Indian Heritage of Knowledge:** Ancient Indian Knowledge: The *Vedas* and its components-the *Vedangas* Ancient Indian books and treaties: The *Sastras*.; The Great Indian Epics: The Ramayana and The Mahabharata Epics and religious treaties of ancient Assam: Introduction to Madhav Kandali's *Ramayan* and Srimanta Sankardev's *Dasam Skandha Bhagavat* of the Puranas; Ancient Traditional Knowledge-The *Agamas ;* The ancient Buddhist knowledge: *Tripitaka: Vinaya, Sutta* and *Abhidhamma Pitaka* **Languages and language studies in India**: What is linguistics?; Script and Language; Alphabet of the Indian; languages *Varnamala:* Origin, Evolution, and phonetic features; Languages of India; Important texts of Indian languages: Skills *Siksha,* Expression/Prounciation-*Nirukta,* Grammer-*Vyakarana,* Poetic rhythm- *Chandas;*Paninian Grammar: A Brief Introduction **Introduction to Fine Arts and Performing Arts of India:** Ancient Indian classical music and dance forms: The Science of Dramas*Natyasastra* and the Science of Music-*Gandharva-Veda;* Aesthetics in Indian Art and Culture; Folk music and traditional dance forms of the Northeast.	15
III	**Indian Science & Technology:** Ancient India's contribution to Mathematics - Number System. Algebra and Arithmetic, Geometry and Trigonometry; Origin of Decimal system in India; nomenclature of numbers in the Vedas. Zero and Infinity. Sulba-sutras. Contribution of Brahmagupta and Sridhar Acharya to Mathematics. Important texts of Indian mathematics. **Indian Astronomy:** Planetary System. Motion of the Planets; Velocity of Light; Eclipse. Astronomy. Navagrahas. Important works in Indian Astronomy.	15

	Aryabhata and Nilakantha: Contribution to Astronomical Studies **Indian Metal Works:** Mining Techniques. Types of Metals. Tools & Techniques for Metal Smelting with examples. Metalworks in premodern India: Special reference to NE India.	
IV	**Contribution of Ancient India to Health Sciences:** Traditional Indigenous systems of medicines in India: - *Ayurveda* and *Yoga;* Elements of *Ayurveda: Gunas* and *Doshas*, *Pancha Mahabhuta* and *Sapta-dhatu;* Concept of disease in Ayurveda; Ayurvedic lifestyle practices: *Dinacharya* and *Ritucharya;* Important Ayurvedic Texts; Hospitals in Ancient India; *Ayurveda*: Gift of India to the modern world.	15
Total		**60**

The experiential learning sessions may include:
• Field Visits: Organizing visits to historical sites, museums, traditional craft centers, and other places relevant to Indian knowledge systems. • Interactive Sessions: Engaging students in discussions with experts and practitioners in various fields of Indian knowledge systems to gain insights and practical knowledge. • Online Lecture Series: Providing the students with online lectures by distinguished experts in the field of the Indian Knowledge System. • Hands-on Activities: Providing opportunities for students to participate in activities related to traditional arts, crafts, music, dance, agriculture, etc., to understand the practical aspects of Indian knowledge systems. • Practical Demonstrations: Conducting workshops or sessions to demonstrate traditional practices, such as yoga, Ayurveda, Vastu Shastra, etc., for the students.

Textbooks Books:

1. Mahadevan, B., Bhat Vinayak Rajat, Nagendra Pavan RN. (2022), *Introduction to Indian Knowledge System: Concepts and Applications.* PHI Learning Private Ltd.
2. Mukul Chandra Bora*, Foundations of Bharatiya Knowledge System.* Khanna Book Publishing

Reference Books:

1. Baladev Upadhyaya, *Samskrta Śāstrom ka Itihās*, Chowkhambha, Varanasi, 2010.

2. D. M. Bose, S. N. Sen and B. V. Subbarayappa, Eds., *A Concise History of Science in India*, 2nd Ed., Universities Press, Hyderabad, 2010.
3. Astāngahrdaya, Vol. I, *Sūtrasthāna and Śarīrasthāna*, Translated by K. R. Srikantha Murthy, Vol. I, Krishnadas Academy, Varanasi, 1991.
4. Dharampal, *The Beautiful Tree: Indian Indigenous Education in the Eighteenth Century*, Dharampal Classics Series, Rashtrotthana Sahitya, Bengaluru, 2021.
5. J. K. Bajaj and M. D. Srinivas, *Indian Economy, and Polity in Eighteenth-century Chengalpattu*, in J. K. Bajaj ed., Indian Economy and Polity, Centre for Policy Studies, Chennai, 1995, pp. 63-84.

Type of Course: AEC
(w.e.f. 2023-24) UG programmes Semester:
1st Course Code:
CEN982A101
Course Title: CEN I: Introduction to Effective Communication Total credits: 1
Course level: 100

L-T-P-C: 1-0-0-1

Scheme of Evaluation: Theory and Practical

Course Objective: To understand the four major aspects of communication by closely examining the processes and figuring the most effective ways to communicate with interactive activities.

Course Outcomes: On successful completion of the course the students will be able to

SI No	Course Outcome	Blooms Taxonomy Level
CO 1	**Identify** the elements and processes that make for successful communication and **recognise** everyday activities that deserve closer attention in order to improve communication skills	**BT 1**
CO 2	**Contrast** situations that create barriers to effective communication and **relate** them to methods that are consciously devised to overcome such hindrance	**BT 2**
CO 3	**Use** language, gestures, and para-language effectively to avoid miscommunication and **articulate** one's thoughts and build arguments more effectively	**BT 3**

Detailed Syllabus		
Units	**Course Contents**	**Periods**
I	Introduction to Effective Communication • Listening Skills o The Art of Listening o Factors that affect Listening o Characteristics of Effective Listening • Guidelines for improving Listening skills	**5**

II	• Speaking Skills o The Art of Speaking o Styles of Speaking o Guidelines for improving Speaking skills o Oral Communication: importance, guidelines, and	**5**
	barriers	
III	• Reading Skills o The Art of Reading o Styles of Reading: skimming, surveying, scanning • Guidelines for developing Reading skills	**5**
IV	• Writing Skills o The Art of Writing o Purpose and Clarity in Writing o Principles of Effective Writing	**5**

Keywords: Communication**,** Listening, Speaking, Reading, Writing

Text:

1. *Business Communication* by Shalini Verma

References:

1. *Business Communication* by P.D. Chaturvedi and Mukesh Chaturvedi
2. *Technical Communication* by Meenakshi Raman and Sangeeta Sharma

Credit Distribution		
Lecture/Tutorial	**Practicum**	**Experiential Learning**
15 hours	-	10 hours – Movie/ Documentary screening – Peer teaching – Seminars – Field Visit

Subject Name: Behavioural Sciences -1
UG 1st semester
Course code: BHS982A102
Credit: 1

Course objectives: To increase one's ability to draw conclusions and develop inferences about attitudes and behaviour, when confronted with different situations that are common in modern organizations.

Course Outcomes: On completion of the course the students will be able to :

CO1: Understand self & process of self exploration- BT 1

CO2: Learn about strategies for development of a healthy self-esteem- BT 2 CO3: Apply the concepts to build emotional competencies-BT 3

Detailed Syllabus:

Modules	Course Contents	Periods
I	**Introduction to Behavioral Science** Definition and need of Behavioral Science, Self: Definition components, Importance of knowing self, Identity Crisis, Gender and Identity, Peer Pressure, Self-image: Self Esteem, Johari Window, Erikson's model.	4
II	**Foundations of individual behavior** Personality- structure, determinants, types of personalities. Perception: Attribution, Errors in perception. Learning- Theories of learning: Classical, Operant and Social	4
III	**Behaviour and communication.** Defining Communication, types of communication, barriers to communication, ways to overcome barriers to Communication, Importance of Non-Verbal Communication/Kinesics, Understanding Kinesics, Relation between behaviour and communication.	4
IV	**Time and Stress Management** Time management: Introduction-the 80:20, sense of time management, Secrets of time management, Effective scheduling. Stress management: effects of stress, kinds of stress-sources of stress, Coping Mechanisms. Relation between Time and Stress.	4
Total		**16**

Text books

- J William Pfeiffer (ed.) Theories and Models in Applied Behavioural Science, Vol 3, Management; Pfeiffer &Company
- Blair J. Kolasa, Introduction to Behavioural Science for Business, John Wiley & Sons Inc
- K.Alex, Soft skills; S.Chand.

SEC

Paper SEC Course	Basics of Cartography L-T-P-C: 0-0-6-3 Scheme of Evaluation: (P) Course Level : 100	Subject Code: GEO162S111

Course Objectives: *This course primarily focuses on the basic concepts of the art and science of Cartography and map making in geographical studies. It also aims to develop skills in spatial representation, data visualization, and technological applications in mapping, thereby promoting innovation, sustainable infrastructure planning, and informed decision-making in alignment with SDG 9: Industry, Innovation and Infrastructure.*

Course Outcomes:

After successful completion of the course, the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Recall** knowledge regarding the classification, elements, and technological significance of maps in geographical analysis and infrastructure planning.	BT1
CO2	**Interpret** graphs and prepare qualitative and quantitative thematic maps using innovative cartographic techniques and spatial data representation methods.	BT2
CO3	**Apply** maps and cartographic knowledge for proper utilization in developmental planning, sustainable infrastructure, and regional development processes in relation to SDG 9: Industry, Innovation and Infrastructure.	BT3
CO4	**Examine** the preparation of various thematic maps with the application of different cartographic and geospatial techniques for effective planning and decision-making.	BT4

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Cartography: meaning, trends of development and importance in geography; Map: definition, types, basic characteristics and map symbols; Thematic Mapping: concept and characteristics of Isopleth, Choropleth and Chorochromatic.	15
Unit 2	Concept and types of Map Scale; conversion of map scale from one to another; construction of graphical scale; representation of data through pie graph, sphere graph, bar graph and line graph.	15
Unit 3	Interpretation of topographical maps in terms of physical and cultural features; construction of transect chart.	15

Unit 4	Map Projection: definition, classification and principles of construction- Zenithal Polar, Gnomonic Projection, Gall's Projection, Stereographic Cylindrical Projection and Simple Conical Projection with one standard parallel; Mapping of slope by Wentworth's method; Profile Drawing: Serial, Superimposed, Composite and Projected.	15
	Total	**60**

Notional Credit Hours for the course: 30X3= 90

Total credit in the paper	Lecture/Tutorial	Practicum	Experiential Learning
3	-	90 hours	-

Text Books:

1. Cuff J. D. and Mattson M. T., 1982: ***Thematic Maps: Their Design and Production***, Methuen Young Book
2. Dent B. D., Torguson J. S., and Holder T. W., 2008: ***Cartography: Thematic Map Design (6th Edition)***, Mcgraw-Hill Higher Education

Reference books:

1. Gupta K. K. and Tyagi V. C., 1992: Working with Maps, Survey of India, DST, New Delhi. 4.
2. Kraak M.-J. and Ormeling F., 2003: Cartography: Visualization of Geo-Spatial Data, Prentice-Hall
3. Mishra R. P. and Ramesh A., 1989: Fundamentals of Cartography, Concept, New Delhi.
4. Tyner J. A., 2010: Principles of Map Design, The Guilford Press
5. Singh R. L. and Singh R. P. B., 1999: Elements of Practical Geography, Kalyani Publishers
6. Sarkar, A., 2000: Practical Geography: A Systematic Approach, Orient BlackSwan, India.
7. Monkhouse, F. J., and Wilkinson, H. R., 2022: Maps and Diagram: Their Compilation and Construction, Alphanumera Publisher

Semester 2

B.A/B. Sc. (Honours) Course in Geography: Semester-II

Major (Course)

Paper I Core Course	**CLIMATOLOGY AND OCEANOGRAPHY** **L-T-P-C: 3-0-0-3 Scheme of Evaluation: T Course Level : 200**	**Subject Code: GEO162M-201**

Course Objectives: *The course aims to illustrate atmospheric elements, processes, and resultant weather and climates, along with the impact of climate on planet Earth, oceanic processes, ocean floor topography, and marine resources. It also seeks to develop awareness regarding climate variability, marine ecosystem conservation, and sustainable use of ocean resources in alignment with SDG 13: Climate Action and SDG 14: Life Below Water.*

Course Outcomes:

After successful completion of the course, the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Define** the elements of weather and climate and their impacts at different spatial scales with reference to climate variability and environmental sustainability.	BT1
CO2	**Demonstrate** weather charts, hythergraphs, and other practical exercises related to atmospheric and oceanic studies for understanding climatic patterns and marine conditions.	BT2
CO3	**Develop** an understanding of climatic aspects and their bearing on planet Earth, including oceanic processes, marine resources, and their relevance to SDG 13: Climate Action and SDG 14: Life Below Water.	BT3
CO4	**Distinguish** between different climatic and oceanic phenomena and analyse their influence on marine ecosystems, weather systems, and global environmental change.	BT4

Detailed Syllabus:

	Topics and Course Content	
Unit 1	The structure and composition of Earth's atmosphere; Elements of weather and climate; Factors affecting the distribution of temperature; Vertical and horizontal and seasonal distribution of temperature; Insolation and heat budget; Temperature inversion; Atmospheric pressure and circulation of planetary winds; Air masses and their characteristics	15
Unit 2	Cyclones: Tropical Cyclones, Temperate Cyclones, Monsoon - Origin and Mechanism, Jet Streams; Atmospheric Moisture: Evaporation, Humidity, Condensation, Fog and Clouds, Precipitation Types, Stability and Instability; Climatic Regions, Climate change and global warming, El Nino	15

Unit 3	Ocean floor topography of Indian, Atlantic and Pacific oceans; Oceanic water Movements: Waves, Currents, Tsunamis and Tides; rainfall frequency analysis, water deficiency and surplus graph, weather chart interpretation,	15
Unit 4	Salinity and Temperature of ocean water: Distribution and Determinants; Coral Reefs , Marine Deposits and Ocean Resources.	15
	Total	**60**

Notional Credit Hours for the course: 30X3= 90

Total credit in the paper	Lecture/Tutorial	Experiential Learning
3	60 hours	30 hours
		Field visit to weather stations for micro-climatic study

Text Books:

1. Critchfield, H. J., 1987: ***General Climatology***, Prentice-Hall of India, New Delhi
2. Lal, D.S., 2001, ***Climatology***, Chaitanya Publishing House, Allahabad
3. Vatal, M., and Sharma, R.C., 2018: Oceanography for Geographers, Surjeet Publications, India.

Reference Books:

1. Anikouchine W. A. and Sternberg R. W., 1973: The World Oceans: An Introduction to Oceanography, Prentice-Hall.
2. Barry, R. G. and Carleton , A. M., 2001: *Synoptic and Dynamic Climatology*, Routledge, UK.
3. Barry, R. G. and Chorley, R. J., 1998: *Atmosphere, Weather and Climate*, Routledge, New York.
4. Batten L. J., 1979: Fundamentals of Meteorology, Prentice-Hall Inc., Englewood Cliffs, New Jersey.
5. Boucher K., 1975: Global Climates, Halstead Press, New York.
6. Garrison T., 1998: Oceanography, Wordsworth Company, Belmont.
7. Gerald S., 1963: General Oceanography: An Introduction, John Willey & Sons, New York.
8. Kershaw S., 2000: *Oceanography: An Earth Science Perspective*, Stanley Thornes, UK.
9. King C. A. M., 1962: *Oceanography for Geographers*, Edward Arnold.
10. Lutgens F. K., Tarbuck E. J. and Tasa D., 2009: The Atmosphere: An Introduction to Meteorology, Prentice-Hall, Englewood Cliffs, New Jersey.
11. Oliver J. E. and Hidore J. J., 2002: Climatology: An Atmospheric Science, Pearson Education, New Delhi.
12. Pinet P. R., 2008: Invitation to Oceanography (Fifth Edition), Jones and Barlett Publishers, USA, UK and Canada.
13. Singh, S. Climatology, 2007, Sharada Pustak Bhawan, Allahabad
14. Strahler, Arthur. N., 1987: *Modern Physical Geography*, John Wiley and Sons, New York,
15. Singapore.
16. Strahler, A., 2018: Introducing *Physical Geography*, John Wiley and Sons, New York, Singapore.

17. Sharma R. C. and Vatal M., 1980: *Oceanography for Geographers*, Chaitanya Publishing House, Allahabad.
18. Trewartha G. T. and Horne L. H., 1980: *An Introduction to Climate*, McGraw-Hill.
19. Thurman H. V., 1996: *Essentials of Oceanography*, Prentice-Hall, New Jersey

Interdisciplinary Course	GEOGRAPHY OF TOURISM L-T-P-C: 3-0-0-3 Scheme of Evaluation: T Course Level : 200	Subject Code: GEO162M-202

Course Objectives: *The course aims to make students understand the basic themes and concepts of Tourism Geography and interpret the geographical components of tourism. It also seeks to promote awareness of sustainable tourism practices, local economic development, and responsible resource utilization in alignment with SDG 8: Decent Work and Economic Growth and SDG 12: Responsible Consumption and Production.*

Learning Outcome:

Course Outcomes:

After successful completion of the course, the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Define** the geographical aspects that organise economic space with reference to tourism, employment generation, and sustainable economic growth.	BT1
CO2	**Illustrate** the geographical aspects of tourism in an area while recognising the importance of sustainable tourism practices and responsible resource management.	BT2
CO3	**Develop** practical field knowledge about tourist places across India and assess their contribution to local livelihoods, cultural preservation, and sustainable development in relation to SDG 8: Decent Work and Economic Growth and SDG 12: Responsible Consumption and Production.	BT3
CO4	**Analyse** the knowledge gathered through field visits and prepare reports focusing on tourism planning, environmental sustainability, and responsible tourism management.	BT4

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Meaning, Scope and contents of Geography of tourism: Importance of geography of tourism; Types of tourism; Motivations of tourism; Components of Tourism;	15
Unit 2	Impacts of tourism: Environmental, Social, Cultural, and Economic impacts of tourism; Definition and principles of sustainable tourism development; Concept of carrying capacity; Concept of Responsible tourism	15
Unit 3	Tourism resources: Destination and resource factors; Mass tourism vs. alternative tourism; Ecotourism; Spatial pattern of Tourism Resources in India-National Parks, Wildlife sanctuaries, Tiger Reserves, Biosphere reserves & wetlands, history and culture	15

Unit 4	Major Tourist attractions in India and abroad, Leiper's tourism model, Current trends of tourism at global scale. Visit to place of sustainable tourism significance. On completion of the tour, the students have to submit a tour report along with an oral presentation.	15
	Total	**60**

Notional Credit Hours for the course: 30X3= 90

Total credit in the paper	Lecture/Tutorial	Experiential Learning
3	60 hours	30 hours
		Field visit to tourism sites and study of sustainable tourism management

Text Books:

1. Bhatia, A. K., 1996***: Tourism Development: Principles and Practices,*** Sterling Publishers, New Delhi.
2. Sharma J. K. (ed.), 2000: ***Tourism Planning and Development - A new perspective***, Kanishka Publishers, New Delhi.

Reference Books:

1. Robinson, H., 1996: ***A Geography of Tourism***. Macdonald and Evans, London, 1996.
2. Williams Stephen, 1998: ***Tourism Geography***, Routledge, Contemporary Human
3. Geography Series, London.
4. Shaw G. and Williams A. M., 1994: ***Critical issues in Tourism-A Geographical Perspective***, Oxford: Blackwell

Minor

Paper I Core Course	**Fundamentals of Human Geography** **L-T-P-C: 3-0-0-3 Scheme of Evaluation: T Course Level : 200**	**Subject Code: GEO162N201**

Course Objectives: *The objective of this course is to make students understand the chronological development of Human Geography through the contributions of various scholars, approaches, and schools of Human Geography, along with the major themes and components of Cultural Geography. The course also aims to foster awareness regarding social inclusion, cultural diversity, urban sustainability, and spatial justice in alignment with SDG 10: Reduced Inequalities and SDG 11: Sustainable Cities and Communities.*

Course Outcomes:

After successful completion of the course, the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Define** the various parameters and components of Human Geography with reference to social diversity, cultural patterns, and spatial inequalities.	BT1
CO2	**Interpret** the development of a humanistic view of Geography in the context of inclusive societies, community well-being, and sustainable human settlements.	BT2
CO3	**Identify** the various aspects of Human Geography and their relevance to urban sustainability, cultural integration, and equitable development in relation to SDG 10: Reduced Inequalities and SDG 11: Sustainable Cities and Communities.	BT3
CO4	**Discover** the humanistic perspective and its dimensions in Geography in relation to physical and cultural surroundings while analysing issues of inequality, cultural identity, and sustainable communities.	BT4

Detailed Syllabus:

Modules	**Topics and Course Content**	**Periods**
Unit 1	Meaning, subject matter and Scope of Human Geography, human versus physical geography; branches of Human Geography; Development of Human Geography; Contributions of German and French Geographers. Man-environment relationship, Determinism vs possibilism,	15
Unit 2	World population distribution and growth, components of population growth: fertility, mortality, migration, : Basic concepts of population: under population, optimum population, over population, Concept of population explosion and population pressure,	15
Unit 3	Population composition: age, sex, child women ratio and other related factors, Population policies in developed and developing economies, Population resource relationship, population-resource regions	15

Unit 4	Man and culture, Characteristics of culture, its components and functions; Cultural diffusion and factors associated with it, concept of cultural landscape, Major cultural regions of the World and their characteristics.	15
	Total	**60**

Notional Credit Hours for the course: 30X3= 90

Total credit in the paper	Lecture/Tutorial	Experiential Learning
3	60 hours	30 hours
		Field work and report making

Text Books:

1. Huntington, E., 1951: ***Principles of Human Geography***, John Wiley & Sons, Inc, New York
2. Hussain, M., 1994: ***Human Geography***, Rawat Publication, New Delhi.

Reference books:

3. Haggett, P., 1972: Geography: A Modern Synthesis, Harper & Row, New York
4. Singh, S., 1991: Environmental Geography, Pustak Bhawan, Allahabad
5. Strahler, A.N. & A.H. Strahler, 1976: Geography and Man's Environment, John Willey, New York

Semester – II		
Paper II/Subject Name: Introduction to Indian Knowledge System - II **Subject Code: IKS992K201** **L-T-P-C – 2-1-0-3** **Credit Units: 3** **Course Level: 100** **Scheme of Evaluation: Summative Assessment - 70% + Formative Assessment -30%**		
Credit Distribution (NCH)		
Lecture/Tutorial (Hours)	**Practical (Hours)**	**Experiential Learning (Hours)**
60	0	30

Course objectives:

This Foundation course is designed to present an overall introduction to all the streams of IKS relevant to the UG program. It would enable students to explore the most fundamental ideas that have shaped Indian Knowledge Traditions over the centuries.

Course Outcomes:

On completion of this course, students will be expected to –

CO	Contents	BT Level
CO_1	Recall traditional Indian knowledge traditions constituting Indian culture	BT level 1
CO_2	Summarize differences between classical literature in Sanskrit and other Indian languages	BT level 2
CO_3	Compare knowledge traditions originating in NE India	BT level 2
CO_4	Appreciate the contribution of Indian Knowledge Systems to the world	BT level 3

Module	Course Contents	Periods
I	**Indian Classical Literature** Indian Classical Literature: A Brief Introduction; Ancient Indian Spritual Poetics-*Kavya*: Contribution of Kalidasa **Diversity and Indian Culture**: Diversity and Indian Culture; Indigenous Faith and Religion; Preservation of culture and indigenous knowledge **The Purpose of Knowledge:** Understanding Self-Awareness and Spirituality; Indian concept and purpose of Knowledge and Education; Understanding Spirituality and Materialism: *Para* and *Apara Vidya*	15
Module	**Course Contents**	**Periods**
II	**Methodology of Indian Knowledge System**: *Shruti* and *Smriti* traditions; Intoduction to *Shastras;* Manuscriptology: The art and science of documenting knowledge; Repositories of ancient manuscripts with special reference to the Norteast India. **Indian Architecture and Town Planning**: Introduction ancient Indian architecture; *Sthapatya-Veda*: An Introduction; Indigenous tools & techniques for town planning & Temple Architecture. Lothal, Mohan Jo Daro; Temple Art: Lepakshi Temple, Jagannath Puri Temple, Konark Sun Temple; Vernacular architecture of Assam: Special reference to Brahmaputra Valley	15
III	**Indian Agriculture**: Agriculture: Significance in Human Civilization; Sustainable Agriculture; Historical significance of agriculture and sustainable farming in India;Step Cultivation of India: Special reference to Northeast India; Wet rice cultivation of Assam. **Indian Textiles**: What is Textile?; Tradition of cotton and silk textiles in India; The historical contribution of textile and weaving to the Indian economy; Varieties of textiles and dyes developed in different regions of India with special reference to Northeast India	15
IV	**Indian Polity and Economy**: Understanding Kingdom and Chiefdom; Role of a king; The Indian idea of a well-organized polity and flourishing economy; The *Chakravarti* System: Administrative System of Ancient Bharatvarsha; Village administrative system: Northeast India; *Arthashastra*: Brief synopsis **The outreach of Indian Knowledge System across Geographical Boundaries:** Indian Languages; Scripts; Linguistics; Ayurveda; Yoga and Meditation; Textile; Decimal value place system-based arithmetic, Algebra and Astronomy	15
Total		**60**

The experiential learning sessions may include:
• Field Visits: Organizing visits to historical sites, museums, traditional craft centers, and other places relevant to Indian knowledge systems. • Interactive Sessions: Engaging students in discussions with experts and practitioners in various fields of Indian knowledge systems to gain insights and practical knowledge. • Online Lecture Series: Providing the students with online lectures by distinguished experts in the field of the Indian Knowledge System. • Hands-on Activities: Providing opportunities for students to participate in activities related to traditional arts, crafts, music, dance, agriculture, etc., to understand the practical aspects of Indian knowledge systems. • Practical Demonstrations: Conducting workshops or sessions to demonstrate traditional practices, such as yoga, Ayurveda, Vastu Shastra, etc., for the students.

Textbooks Books:

3. Mahadevan, B., Bhat Vinayak Rajat, Nagendra Pavan RN. (2022), *Introduction to Indian Knowledge System: Concepts and Applications.* PHI Learning Private Ltd.
4. Mukul Chandra Bora*, Foundations of Bharatiya Knowledge System.* Khanna Book Publishing

Reference Books:

6. Baladev Upadhyaya, *Samskrta Śāstrom ka Itihās*, Chowkhambha, Varanasi, 2010.
7. D. M. Bose, S. N. Sen and B. V. Subbarayappa, Eds., *A Concise History of Science in India*, 2nd Ed., Universities Press, Hyderabad, 2010.
8. Asta¯ ngahrdaya, Vol. I, *Sūtrasthāna and Śarīrasthāna*, Translated by K. R. Srikantha Murthy, Vol. I, Krishnadas Academy, Varanasi, 1991.
9. Dharampal, *The Beautiful Tree: Indian Indigenous Education in the Eighteenth Century*, Dharampal Classics Series, Rashtrotthana Sahitya, Bengaluru, 2021.
10. J. K. Bajaj and M. D. Srinivas, *Indian Economy, and Polity in Eighteenth-century Chengalpattu*, in J. K. Bajaj ed., Indian Economy and Polity, Centre for Policy Studies, Chennai, 1995, pp. 63-84.

Type of Course: AEC (w.e.f. 2023-24)
UG programmes Semester: 2nd Course Code: CEN982A201
Course Title: CEN II: Approaches to Verbal and Non-Verbal Communication
Total credits: 1 **Course level: 100** **L-T-P-C: 1-0-0-1**

Course Objectives

To introduce the students to the various forms of technical communication and enhance their knowledge in the application of both verbal and non-verbal skills in communicative processes. **Course Outcomes**

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Identify** the different types of technical communication, their characteristics, their advantages and disadvantages.	**BT 1**
CO 2	**Explain** the barriers to communication and ways to overcome them.	**BT 2**
CO 3	**Discover** the means to enhance conversation skills.	**BT 3**
CO 4	**Determine** the different types of non-verbal communication and their significance.	**BT4**

Detailed Syllabus

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I	**Technology Enabled Communication** Communicating about technical or specialized topics, Different forms of technology-enabled communication tools used in organisations Telephone, Teleconferencing, Fax, Email, Instant messaging , Blog, podcast, Videos, videoconferencing, social media	**4**
II	**Communication Barriers** Types of barriers: Semantic, Psychological, Organisational, Cultural, Physical, and Physiological. Methods to overcome barriers to communication.	**4**
III	**Conversation skills/Verbal Communication** Conversation – Types of Conversation, Strategies for Effectiveness, Conversation Practice, Persuasive Functions in Conversation, Telephonic Conversation and Etiquette Dialogue Writing, Conversation Control.	**4**
IV	**Non-verbal Communication** Introduction; Body language- Personal Appearance, Postures, Gestures, Eye Contact, Facial expressions Paralinguistic Features-Rate, Pause, Volume, Pitch/Intonation/ Voice/ modulation Proxemics , Haptics, Artifactics, Chronemics	**4**
	Total	**16**

Texts:

1. Rizvi, M. Ashraf. (2017). *Effective Technical Communication*. McGraw-Hill.

2. Chaturvedi, P. D. and Chaturvedi, Mukesh. (2014). *Business Communication.*Pearson.
3. Raman, Meenakshi and Sharma, Sangeeta. (2011). *Technical Communication: Principles and Practice* (2nd Edition): Oxford University Press.

References:

1. Hair, Dan O., Rubenstein, Hannah and Stewart, Rob. (2015). *A Pocket Guide to Public Speaking.* (5th edition). St. Martin's. ISBN-13:978-1457670404
2. Koneru, Aruna.(2017) *Professional Communication*. New Delhi: Tata McGraw Hill ISBN-13: 978-0070660021
3. Raman, Meenakshi and Singh, Prakash.(2012). *Business Communication* (2nd Edition): Oxford University Press
4. Sengupta, Sailesh.(2011) *Business and Managerial Communication*. New Delhi : PHI Learning Pvt. Ltd.

Subject Name: Behavioural Sciences -II **UG 2nd Semester**

Course code: BHS982A202

Credit: 1

Course objectives: To increase one's ability to draw conclusions and develop inferences about attitudes and behaviour, when confronted with different situations that are common in modern organizations.

Course outcomes: On completion of the course the students will be able to:

CO 1: Develop an elementary level of understanding of culture and its implications on personality of people-BT 1

CO2: Understand the concept of leadership spirit and to know its impact on performance of employees- BT 2

CO3: Understand and apply the concept of Motivation in real life- BT 3

Modules	Course Contents	Periods
I	**Culture and Personality** Culture: Definition, Effect, relation with Personality, Cultural Iceberg, Overview of Hofstede's Framework, Discussion of the four dimensions of Hofstede's Framework.	4
II	**Attitudes and Values** Attitude's definition: changing our own attitudes, Process of cognitive dissonance Types of Values, Value conflicts, Merging personal and Organisational values	4
III	**Motivation** Definition of motivation with example, Theories of Motivation (Maslow, McClelland's theory& Theory X and Y)	4
IV	**Leadership** Definition of leadership, Leadership continuum, types of leadership, Importance of Leadership, New age leaderships: Transformational & transactional Leadership, Leaders as role models.	4
Total		**16**

Text books:

- J William Pfeiffer (ed.) Theories and Models in Applied Behavioural Science, Vol 3, Management; Pfeiffer &Company
- Blair J. Kolasa, Introduction to Behavioural Science for Business, John Wiley & Sons Inc.
- Organizational Behaviour by Kavita Singh (Vikas publishers, 3rd Edition).

SEC

SEC	FUNDAMENTALS OF GEOINFORMATICS L-T-P-C: 0-0-6-3 Scheme of Evaluation: P Course Level : 200	Subject Code: GEO162S211

Course Objectives: *The course aims to make students interpret data, tools, technologies, and applications of Geoinformatics, including GIS, Remote Sensing, and GPS, and to construct and analyse maps using Geospatial Technology. It also seeks to promote technological innovation, sustainable infrastructure planning, and data-driven decision-making in alignment with SDG 9: Industry, Innovation and Infrastructure.*

Course Outcomes:

After successful completion of the course, the students will be able to:

CO	Course Outcome	Blooms Taxonomy Level
CO1	**Define** the fundamental terms and terminologies of Geoinformatics and their significance in technological and infrastructural development.	BT1
CO2	**Outline** the strengths and applications of Geospatial Technology in resource management, spatial planning, and innovative infrastructure development.	BT2
CO3	**Build** maps of resources, their locations, and availability using geospatial techniques for sustainable planning and development in relation to SDG 9: Industry, Innovation and Infrastructure.	BT3
CO4	**Analyse** different remote sensing datasets collected from various platforms for spatial analysis, infrastructure planning, environmental monitoring, and informed decision-making.	BT4

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Geoinformatics- Meaning and scope, The Earth: shape, size, and earth models; Referencing systems; Definition of map, map properties, Geospatial data types and structure and their characteristics; Georeferencing a scanned map, Creation of vector data layers - point, line and polygon and map lay;	15
Unit 2	Vector data editing; Vector Attribute database preparation; Basics of spatial and non-spatial / attribute database, relational database; Attribute mapping / thematic mapping of various attributes of point, line and polygon attributes	15
Unit 3	Basic concept of Remote Sensing; Satellites – geostationary and remote sensing Satellites; Sensors- Types and characteristics; Resolution- Types and importance (spatial, spectral, radiometric and temporal) Global Position System (GPS) and its applications.	15
Unit 4	Applications of Geoinformatics: Urban planning and management, Environmental monitoring and management, Natural resource management, Disaster management, Transportation planning, Agriculture and forestry, Public health.Ethical, Legal, and Social	15

	Issues in Geoinformatics: Privacy concerns. Data ownership and copyright issues, Security considerations Ethical use of geospatial data	
	Total	**60**

Notional Credit Hours for the course: 30X3= 90

Total credit in the paper	Lecture/Tutorial	Experiential Learning	Total credit in the paper
3	60 hours	30 hours	3
		Hands on practice in Lab and assignment preparation	

Text Books:

1. De Mars, M. N., 1999: **Fundamentals of Geographic Information Systems**, John Wiley & Sons Inc., New York.
2. Jensen, J. R., 2011: **Remote Sensing of the Environment – An Earth Resource Perspective**, 3rd Impression, Pearson, New Delhi.

Reference Books:

1. Burrough, P.A. and Mc Donnel, R. A., 1998: **Principles of Geographical Information Systems**, Oxford University Press.
3. Chetry, N., 2019 (Ed): A Glimpse of Geospatial Technology and Applications, Eastern Book House, Guwahati
4. Sabins, Floyd F., 1987: **Remote Sensing Principles and Interpretation**, W.H. Freeman and Company, New York.
5. Chang, K. T., 2018: Introduction to Geographic Information Systems, McGraw-Hill Education, New York.

NOTE: Software packages: ArcGIS / QGIS /ILWIS, ERDAS Imagine/SAGA/ILWIS. Record of the exercises (duly signed by the teacher concerned within specified date for each exercise) in the form of Practical Note Book to be made by the students is mandatory. Records of all exercises of each unit are compulsory. Maximum 20-25% marks of Semester End Examination may be kept for viva and practical note book with equal weightage on each aspect. Rest of the marks may be more or less / equally distributed to each exercise depending on difficulty level. Semester End Examination will be of 4 (four) hours duration with maximum 2 (two) exercises only from any of the units.

VAC Course	GEOGRAPHY OF INDIA L-T-P-C: 4-0-0-4 Credit Units: 4 (T) Scheme of Evaluation:	Subject Code: VAC992V2015

Course Objectives: *The course aims to define the regional basis of India and evaluate the fundamental aspects of the country, including its physical, economic, social, and developmental characteristics. It also seeks to create awareness regarding food security, sustainable agriculture, energy resources, and regional development in alignment with SDG 2: Zero Hunger and SDG 7: Affordable and Clean Energy.*

Course Outcomes:

By the end of this course, the students will be able to:		
CO	**Course Outcome**	**BloomsTaxonomy Level**
CO1.	**Define** the concepts involved in explaining India as a regional unit with.	**BT 1**
CO2.	**Compare** and **interpret** the disparities prevailing among different states of India in terms of food production, energy access, and regional development.	**BT 2**
CO3.	**Build** knowledge on population structure, industrial aspects, transport, communication, agriculture, and energy resources of the region in relation to SDG 2: Zero Hunger and SDG 7: Affordable and Clean Energy.	**BT 3**
CO4.	**Analyse** the various developmental prospects of India with emphasis on sustainable agriculture, food security, renewable energy, and balanced regional growth.	**BT 4**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	India as a geographical entity, Location and situation; India in the context of neighbouring counties, Physical background of regional development: relief, drainage system, climate, soil and natural vegetation, major crops, Indian monsoon: mechanism and characteristics, natural disaster in India (earthquake, drought, flood, cyclone, tsunami, Himalayan highland hazards)	10
Unit 2	Population and development issues: population growth and its socio-economic implications, literacy, urbanization, occupation and social structure and development inequalities, tribes and religion	14
Unit 3	Mineral and power resources and development: iron ore, coal, petroleum and hydro- electric power potential, agro-climatic and physiographic divisions of India, agro climatic regions, industrial regions of India	12
Unit 4	Regional disparities in economic development: Agriculture, industry and transport and communication, India's geo-economic position in Asia and the world; its economic development policies and international relations.	12
	Total	**48**

Text Books:

1. Singh, R. L., (ed), 1971: India: A Regional Geography, National Geographical Society of
2. India, Varanasi.
3. Bhatt, L. S., 1973: Regional Planning in India, Statistical Publishing Society, Calcutta.
4. Tirtha R. & Gopal Krishna, 1996: Emerging India Reprinted by Rawat Publications, Jaipur.

Reference Books:

1. Dreze, Jean & Amartya Sen (ed.), 1996: India Economic Development and Social opportunity, Oxford University Press, New Delhi.
2. Kundu A. Raza Moonis, 1982: Indian Economy: the Regional Dimension. Spectrum
3. Publishers, New Delhi.
4. Robinson, Francis, 1989 : The Cambridge Encyclopaedia of India, Pakistan, Bangladesh, Sri Lanka, Nepal, Bhutan & Maldives. Cambridge University Press, London.

Semester 3

MAJOR	ECONOMIC GEOGRAPHY L-T-P-C: 4-0-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEO162 M301

Course Objectives: *This course focuses on the basic concepts of Economic Geography and the associated patterns and processes of the primary economic activities of the world. It also aims to develop an understanding of sustainable economic development, industrial growth, innovation, and infrastructural advancement in alignment with SDG 8: Decent Work and Economic Growth and SDG 9: Industry, Innovation and Infrastructure.*

Course Outcomes:

After successful completion of the course, the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Define** the concepts and ways in which geographical aspects organise economic space with reference to employment, industrialization, and regional development.	BT1
CO2	**Compare** different sectors of the economy and arrive at logical conclusions regarding the importance of each sector in national economic development, innovation, and infrastructure growth.	BT2
CO3	**Identify** the principles and significance of Economic Geography in promoting sustainable economic activities, industrial development, and regional planning in relation to SDG 8: Decent Work and Economic Growth and SDG 9: Industry, Innovation and Infrastructure.	BT3
CO4	**Discover** new insights on the relevance of economy and geography and analyse the associated contemporary challenges related to employment, industrialization, infrastructure, and sustainable development.	BT4

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Meaning and scope of Economic Geography; Approaches in Economic Geography; Concept and classification of economic activity; factors influencing economic activities.	10
Unit 2	Primary Activities: Subsistence and Commercial agriculture, forestry, fishing and mining; Secondary Activities: Manufacturing (Cotton Textile, Iron and Steel), Concept of Manufacturing Regions; Special Economic Zones and Technology Parks; Tertiary Activities: Transport, Trade and Services.	14
Unit 3	Theories of Economic Geography: Von Thunen's model of agricultural location, Industrial location theories of Weber, E.M. Hoover, A. Losch, A. Pred and D. M. Smith; Theories of economic development by Myrdal and Rostow.	14
Unit 4	Economic Geography of Resources; Global pattern of distribution and production of selected resources: Food grains, iron ore, coal, petroleum and nuclear power; Global economic scenario.	10
	Total	**48**

Text Books:

1. Alexander J. W., 1963: ***Economic Geography***, Prentice-Hall Inc., Englewood Cliffs, New Jersey.
2. Coe N. M., Kelly P. F. and Yeung H. W., 2007: ***Economic Geography: A Contemporary Introduction***, Wiley-Blackwell.

Reference Books:

1. Wheeler J. O., 1998: *Economic Geography*, Wiley..
2. Durand L., 1961: *Economic Geography*, Crowell.
3. Bagchi-Sen S. and Smith H. L., 2006: *Economic Geography: Past, Present and Future*, Taylor and Francis.
4. Willington D. E., 2008: *Economic Geography*, Husband Press.
5. Clark, G. L., Feldman, M. P., Gertler, M. S., & Williams, K. (Eds.). (2003). *The Oxford handbook of economic geography*. Oxford University Press

MAJOR	SOIL AND BIOGEOGRAPHY			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEO162M302

Course Objectives: *The course aims to make students understand the fundamental concepts of Soil Geography and Biogeography under various categories. It also seeks to develop awareness regarding ecosystem conservation, biodiversity management, sustainable land use, and environmental sustainability in alignment with SDG 15: Life on Land and SDG 13: Climate Action.*

Course Outcomes:

By the end of this course the students will be able to:		
CO	**Course Outcome**	**BloomsTaxonomy Level**
CO1	**Define** and understand the basic terms and concepts of Soil Geography and Biogeography with reference to ecosystem functioning and environmental sustainability.	BT1
CO2	**Interpret** important environmental issues related to soil degradation, biodiversity loss, and climate change in the context of SDG 15: Life on Land and SDG 13: Climate Action.	BT2
CO3	**Construct** an understanding of the basic properties, morphology, and processes associated with soil and biogeography for sustainable land and ecosystem management.	BT3
CO4	**Analyse** independently various biodiversity conservation, climate resilience, and environmental management issues related to terrestrial ecosystems.	BT4

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Soil geography: meaning and significance; Soil forming factors: Parent material, organic, climatic, topographic, spatio-temporal dimensions; Processes of soil formation and soil development: Physical, biotic and chemical; Soil profile development and soil catena	10
Unit 2	Physical properties of soils: Morphology, texture, structure, water, air, temperature and other properties of soil; chemical properties of soil and soil reaction; pedogenic regimes; podzolization, laterization, calcification and gleization.	12
Unit 3	Forms and functions of ecosystem: Forest, grassland, marine and mountain ecosystem; trophic level, Energy flux in the ecosystem; material Cycles / bio-energy cycles in the terrestrial ecosystem, concept of food chain, food web and ecological pyramid, Environmental ethics and Deep ecology	12
Unit 4	Biogeography: meaning and significance, Approaches in biogeography: evolutionary and ecological, Concept of biodiversity; Conservation of biotic resources; Biodiversity hotspots; State of forest cover changes in India, environment policy of India, National Forest Policy of India, legal framework for biodiversity protection: Brundtland Commission, Kyoto Protocol, Agenda 21, Sustainable Development Goals, Paris Agreement	14
	Total	**48**

Text Books:

1. Hugget, R. J., 1988: Fundamentals of Biogeography. Routledge, London.
2. Bunting, B. T., 1967: The Geography of Soil, Hutchinson, London.
3. Robinson, H., 1982: Biogeography, E.L.B.S., Mc Donald & Evans, London.
4. Sivaperuman, Chandrakasan et al., (2018): Biodiversity and Climate Change Adaptation in Tropical Islands, Academic Press, London.

Reference Books:

1. Barry, C., 1977: Biogeography - An Ecological and Evolutionary Approach, Cox Blackwell, Oxford.
2. Singh, S. 1991: Environmental Geography, Prayag Publications, Allahabad
3. Tivy, J. 1992: Biogeography: A study of Plants in Ecosphere, 3rd edn. Oliver and Boyd, U.S.A.

Minor	BIOGEOGRAPHY			Subject Code:
	L-T-P-C: 3-0-0-3	Credit Units: 3	Scheme of Evaluation: (T)	GEO162N301

Course Objectives: *The course aims to make students understand the fundamental concepts of Biogeography under various categories. It also seeks to develop awareness regarding biodiversity conservation, ecosystem sustainability, and environmental management in alignment with SDG 15: Life on Land.*

Course Outcomes:

After successful completion of the course, the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Define** and understand the basic terms and concepts of Biogeography with reference to ecosystems, biodiversity, and terrestrial environments.	BT1
CO2	**Interpret** important environmental issues related to biodiversity loss, habitat degradation, and ecosystem conservation.	BT2
CO3	**Construct** the basic concepts of Biogeography and their applications in sustainable management of terrestrial ecosystems in relation to SDG 15: Life on Land.	BT3
CO4	**Analyse** independently various biodiversity conservation and environmental management issues affecting life on land and ecological sustainability.	BT4

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Nature, scope and subject matter of biogeography, ecology, ecosystem, trophic level, food chain and energy flow in ecosystem, nutrient cycle	6
Unit 2	Biomes, distribution of plants and animal, ecological regions with special reference to India	10
Unit 3	Biodiversity, biodiversity hot spot, conservation of biodiversity and mechanism: national park, marine national park, wildlife sanctuary, reserve forest, bird sanctuary; with special reference to India	10
Unit 4	Conservation of the environment, important environmental days, EIA, UNFCC, UNCED, environmental policies in India.	10
		36

Text Books:

1. Odum, E.P., 1977: ***Fundamentals of Ecology***
2. Bhattacharya, N.N., 2003, ***Biogeography***, Rajesh Publications, New Delhi.

Reference Books:

1. Lomolino, M. V., Riddle, B. R., Whittaker, R. J. (2017). Biogeography, fifth edition. (5), 730. Sunderland, MA: Oxford University Press.
2. Savindra, S., 2015, Environmental Geography, Pravalika Publications, Allahabad.
3. Anderson: Ecology for Environmental Science.

Course Interdisciplinary	AGRICULTURAL PRACTICES IN INDIA L-T-P-C: 4-0-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Course Code: GEO162I301

Course Objectives: *The course aims to make students understand the role and significance of agriculture in the Indian economy. It also seeks to develop awareness regarding food security, sustainable agricultural practices, and efficient resource utilization in alignment with SDG 2: Zero Hunger and SDG 12: Responsible Consumption and Production.*

Course Outcomes:

By the end of this course the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Recall** the definition, basic concepts, and scope of Agricultural Geography with reference to food production and sustainable agriculture.	BT1
CO2	**Apply** various indices associated with agriculture to understand agricultural productivity, resource use, and sustainable farming practices.	BT2
CO3	**Develop** basic ideas related to the geographical perspectives of agriculture in India with emphasis on food security, responsible production, and sustainable rural development.	BT3
CO4	**Analyse** the fundamental processes associated with the agricultural system of India, including land use, cropping patterns, resource management, and environmental sustainability.	BT4
CO5	**Appraise** the significance of agriculture in the Indian economy in relation to SDG 2: Zero Hunger and SDG 12: Responsible Consumption and Production through sustainable agricultural development and responsible consumption practices.	BT5

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Place of Indian agriculture in global economy, type, characteristics, growth, distribution and development, Agricultural regions of India and their characteristics,	10
Unit 2	Indian agriculture: types, characteristics, growth, distribution and development, Agricultural land use pattern and shifting cropping pattern in India;	10
Unit 3	Regional variation in the levels of agricultural development in India - Food deficit and food surplus regions; nutritional index, sustainability of agricultural practices and production	14
Unit 4	Problems of Indian agriculture: crop and cropping hazards, sustainability of agricultural practices and production; Technological factors in Indian agriculture, Government policy for agricultural policy and planning and agricultural trade in India.	14
	Total	**48**

Text Books:

1. Bayliss Smith, T. P., 1987: The Ecology of Agricultural Systems. Cambridge University Press, London
2. Morgan, W.B. : Agriculture in the Third World - A Spatial Analysis. Westview Press, Boulder, 1978
3. Hussain, M., 2001: Systematic Agricultural Geography, Rawat Publication, Jaipur and New Delhi.

Reference Books:

1. Grigg, D.B., 1978: *Agricultural Systems of the World: An Evolutionary Approach*, Cambridge
2. University Press, Cambridge.
3. Singh, J., 1976: *Agricultural Geography*, Tata McGraw Hill Pub. Co., New Delhi. Sukla, S. P. and
4. Agarwal, A.K.: *Agriculture in Northeast India*.
5. Hussain, M., 2001: Systematic *Agricultural Geography*, Rawat Publication, Jaipur and New Delhi.
6. Mohammad, N. (ed), 1992: *New Dimensions in Agricultural Geography* (in 8 Volumes), Concept Publishing Company, New Delhi.

Type of Course: AEC **(w.e.f. 2023-24) UG programmes Semester: 3rd Course Code:** CEN982A301
Course Title: CEN III – Fundamentals of Business Communication Total credits: 1
Course level: 200
L-T-P-C: 1-0-0-1
Scheme of Evaluation: Theory and Practical

Course Objective: The aim if the course is to develop essential business communication skills, including effective writing, speaking, and interpersonal communication, to enhance professional interactions, collaboration, and successful communication strategies within diverse corporate environments.

Course Outcomes: On successful completion of the course the students will be able to:

SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Define and list business documents using appropriate formats and styles, demonstrating proficiency in written communication for various business contexts.	**BT 1**
CO 2	Demonstrate confident verbal communication skill through persuasive presentations, active listening, and clear articulation to engage and influence diverse stakeholders.	**BT 2**
CO 3	Apply effective interpersonal communication strategies, including conflict resolution and active teamwork, to foster positive relationships and contribute to successful organizational communication dynamics	**BT 3**

Detailed Syllabus		
Units	**Course Contents**	**Periods**
I	**Business Communication: Spoken and Written** • The Role of Business Communication • Classification and Purpose of Business Communication • The Importance of Communication in Management • Communication Training for Managers • Communication Structures in Organizations • Information to be Communicated at the Workplace • Writing Business Letters, Notice, Agenda and Minutes	**5**
II	**Negotiation Skills in Business Communication** • The Nature and Need for Negotiation • Situations requiring and not requiring negotiations • Factors Affecting Negotiation • Location, Timing, Subjective Factors • Stages in the Negotiation Process • Preparation, Negotiation, Implementation • Negotiation Strategies	**5**

III	**Ethics in Business Communication** • Ethical Communication • Values, Ethics and Communication • Ethical Dilemmas Facing Managers • A Strategic Approach to Business Ethics • Ethical Communication on Internet • Ethics in Advertising	**5**
IV	**Business Etiquettes and Professionalism** • Introduction to Business Etiquette • Interview Etiquette • Social Etiquette • Workplace Etiquette • Netiquette	**5**

Text:

1. *Business Communication* by Shalini Verma

References:

1. *Business Communication* by PD Chaturvedi and Mukesh Chaturvedi
2. *Technical Communication* by Meenakshi Raman and Sangeeta Sharma

Credit Distribution		
Lecture/Tutorial	**Practicum**	**Experiential Learning**
15 hours	-	10 hours - Group Discussion - Presentation - Quiz - Case Study

Subject Name: Behavioural Sciences -III

UG 3rd semester

Course code: BHS982A302

Credit: 1

Course objectives: To increase one's ability to draw conclusions and develop inferences about attitudes and behaviour, when confronted with different situations that are common in modern organizations. To enable the students to understand the process of problem solving and creative thinking.

Course outcomes: On completion of the course the students will be able to:

CO1: Understand the process of problem solving and creative thinking-BT 1

CO2: Develop and enhance of skills required for decision-making- BT 2

Modules	Course Contents	Periods
I	**Problem Solving Process** Defining problem, the process of problem solving, Barriers to problem solving(Perception, Expression, Emotions, Intellect ,surrounding environment)	4
II	**Thinking as a tool for Problem Solving** What is thinking: The Mind/Brain/Behaviour Critical Thinking and Learning: -Making Predictions and Reasoning. -Memory and Critical Thinking. - Emotions and Critical Thinking.	4
III	**Creative Thinking** - Definition and meaning of creativity, - The nature of creative thinking :Convergent and Divergent thinking, - Idea generation and evaluation (Brain Storming) - Image generation and evaluation. - The six-phase model of Creative Thinking: ICEDIP model	4
IV	**Building Emotional Competence** Emotional Intelligence – Meaning, components, Importance and Relevance Positive and Negative emotions Healthy and Unhealthy expression of emotions	4
	Total	16

Text books:

- J William Pfeiffer (ed.) Theories and Models in Applied Behavioural Science, Vol 3, Management; Pfeiffer &Company
- Blair J. Kolasa, Introduction to Behavioural Science for Business, John Wiley & Sons In

SEC	INTRODUCTION TO QUANTITATIVE TECHNIQUES	Subject Code:
	L-T-P-C: 0-0-8-4 Credit Units: 3 Scheme of Evaluation: (P)	GEO162S311

Course Objectives: *This paper provides an understanding of the pure and applied nature of Geography along with the key elements of the discipline. It also aims to develop analytical, statistical, and research-oriented skills through the use of quantitative techniques and data analysis, thereby promoting innovation, technological advancement, and inclusive learning in alignment with SDG 9: Industry, Innovation and Infrastructure and SDG 4: Quality Education.*

Course Outcomes:

After successful completion of the course, the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	**Define** the statistical methods and quantitative techniques used in Geography for scientific analysis, research, and informed decision-making.	**BT 1**
CO2.	**Interpret** various methods and techniques of data collection, tabulation, interpretation, and analysis with the application of innovative and technology-based approaches.	**BT 2**
CO3.	**Identify** the importance of data in Geography and its role in research, planning, education, and infrastructural development in relation to SDG 9: Industry, Innovation and Infrastructure and SDG 4: Quality Education.	**BT 3**
CO4.	**Analyse** data through tabulation, sample size determination, and other quantitative methods by handling field-based data for geographical research and sustainable development planning.	**BT 4**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Use of Data in Geography, Significance of Statistical Methods in Geography; Sources of Data, Scales of Measurement (Nominal, Ordinal, Interval, Ratio).	10
Unit 2	Summarization of data: Measures of Central Tendency (Mean, Median and Mode), Dispersion (Standard Deviation, Variance and Coefficient of Variation).	14
Unit 3	Sampling: Purposive, Random, Systematic and Stratified and their utilities in geographical data collection and analysis.	12
Unit 4	Time series analysis techniques: Moving average and Least Squares Methods; Correlation analysis (Spearman's Rank correlation and Karl Pearson's product moment correlation coefficient); Regression analysis in geographic studies (linear and non-linear); regression residual mapping.	12
	Total	**48**

Text Book:

1. Berry B. J. L. and Marble D. F. (eds.): *Spatial Analysis – A Reader in Geography*.
2. Ebdon D., 1977: *Statistics in Geography: A Practical Approach*.

3. Gregory, S. (2014). *Statistical methods and the geographer*. Routledge.
4. Hammond P. and McCullagh P. S., 1978: *Quantitative Techniques in Geography: An Introduction,* Oxford University Press.
5. King L. S., 1969: *Statistical Analysis in Geography*, Prentice-Hall.

References:

1. Mahmood A., 1977: *Statistical Methods in Geographical Studies,* Concept.
2. Monkhouse, F.J. &Wilkinson,H.R.,1989: Maps and Diagrams, B.I Publication, New Delhi
3. Pal S. K., 1998: *Statistics for Geoscientists*, Tata McGraw Hill, New Delhi.
4. Sarkar, A. (2013) Quantitative geography: techniques and presentations. Orient Black Swan Private Ltd., New Delhi
5. Silk J., 1979: *Statistical Concepts in Geography*, Allen and Unwin, London.
6. Spiegel M. R.: Statistics, *Schaum's Outline Series.*
7. Yeates M., 1974: *An Introduction to Quantitative Analysis in Human Geography*, McGraw Hill, New York.

Semester 4

Major	SOCIAL AND POLITICAL GEOGRAPHY			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 3	Scheme of Evaluation: (T)	GEO162M401

Course Objectives: *The course aims to make students understand the basic concepts related to Social and Cultural Geography within the geographical framework and provide knowledge on political systems and geopolitics in the spatial context. It also seeks to promote awareness regarding social justice, gender equality, inclusive societies, and democratic institutions in alignment with SDG 5: Gender Equality and SDG 16: Peace, Justice and Strong Institutions.*

Course Outcomes:

After successful completion of the course, the students will be able to:		
Sl. No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Define** the fundamental concepts of social and political dimensions with reference to equality, justice, and inclusive governance.	BT1
CO2	**Interpret** social, cultural, and political concepts in a broader and analytical manner while understanding issues related to gender, peace, and institutional frameworks.	BT2
CO3	**Build** knowledge on the structures and formations of countries as well as various schools of Political Geography with emphasis on democratic institutions, gender-sensitive governance, and social inclusion in relation to SDG 5: Gender Equality and SDG 16: Peace, Justice and Strong Institutions.	BT3
CO4	**Analyze** socio-cultural and political themes in geographical dimensions with focus on equality, justice, conflict resolution, governance, and inclusive development.	BT4

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Definition and field of social geography; Concept of social differentiation, social structure and social stratification as reflected in race, tribe, caste, language, dialect and religion in India; concept of social well-being and its determination, Concept of Space: Types and characteristics of space.	6
Unit 2	Review of five year plans and area plans towards social policy in India; Strategies to improve social well-being in tribal, hill, drought and flood-prone areas; Spatial distribution of social groups in India	10
Unit 3	Nature, scope and subject matter of political geography; Approaches to the study of political geography, The field and school of thoughts in political geography: landscape school, ecology school	10
Unit 4	Concepts in political geography: frontier and boundary (with reference to India), lebensraum, state and nation, core-periphery and capital, buffer zone, federal state, Colonialism, desalinization, theories of Rim-land and Heartland.	10
	Total	**36**

Text Books:

1. Sen, J (2016): ***A Textbook of Social and Cultural Geography***, Kalyani Publishers, New Delhi
2. Dwiveda R. L. (2019): ***Fundamentals of Political Geography***, Surjeet Publications, Delhi

Reference Books:

1. John R. S., 1982: An introduction to Political Geography, Routledge, London
2. Ahmad, A., 1999: Social Geography, Rawat Publication, Jaipur and New Delhi
3. Ahmad, A. (ed), 1993: Social Structure and Regional Development: A Social Geography perspective, Rawat Publication, Jaipur
4. Pounds N .J. G. (1972): ***Political Geography***, McGraw Hill, New York

MAJOR	ENVIRONMENTAL GEOGRAPHY			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEO1M402

Course Objectives: *The course aims to provide an understanding of the concept of the global environment and its impacts on various aspects of human and natural systems, along with knowledge on climate change adaptation and mitigation strategies and the institutional roles associated with environmental management. The course also promotes awareness regarding ecosystem conservation, climate resilience, and sustainable environmental governance in alignment with SDG 13: Climate Action and SDG 15: Life on Land.*

Course Outcomes:

By the end of this course the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Relate** to the basics of environmental change, climate science, and sustainable development with reference to global environmental challenges.	BT1
CO2	**Classify** different types of natural resources and explain their importance in ecosystem sustainability, biodiversity conservation, and climate resilience.	BT2
CO3	**Develop** understanding about the impacts of climate change on agriculture, water resources, flora and fauna, human health, the ozone layer, and other environmental spheres in relation to SDG 13: Climate Action and SDG 15: Life on Land.	BT3
CO4	**Inspect** issues related to climate adaptation, hazard mitigation, disaster management, and solid waste management for sustainable environmental protection.	BT4
CO5	**Explain** policies of development and environmental protection in developed and developing countries with emphasis on sustainable resource management, biodiversity conservation, and climate governance.	BT5

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Environmental Geography: Concept, Scope and Significance; Human-Environment Relationships: Historical Progression, Adaptation in different Biomes.	14
Unit 2	Eco-system: Concept, types and components, structure and functions; Ecology – Concept and Principles, Environmental Laws in India: Wild life Act, Forest Acts, Environmental Protection Ac.	14
Unit 3	Major Global Environmental Problems: Pollution, Deforestation, Desertification, Global Warming, Bio-Depletion. Management of Environment and Resources; Importance of Environmental Impact Assessment,	10
Unit 4	Environmental Programmes and Policies – Global, National and Local levels; Stockholm Conference, the Earth Summits, Inter- Governmental Panel for Climate Change (IPCC)	10
	Total	**48**

Text Book:

1. Chandna R. C., 2002: *Environmental Geography*, Kalyani, Ludhiana.
2. Miller G. T., 2004: *Environmental Science: Working with the Earth,*

*Thomson*BrooksCole, Singapore.
3. Goudie A., 2001: *The Nature of the Environment*, Blackwell, Oxford.
4. MoEF, 2006: *National Environmental Policy-2006*, Ministry of Environment and Forests, Government of India.

References:

1. Odum, E. P. et al, 2005: *Fundamentals of Ecology*, Ceneage Learning India.
2. Singh S., 1997: *Environmental Geography*, PrayagPustakBhawan, Allahabad.
3. Singh, R.B. (Eds.) (2009) *Biogeography and Biodiversity*. Rawat Publication, Jaipur
4. Singh, R.B. (1998) *Ecological Techniques and Approaches to Vulnerable Environment*, New Delhi, Oxford & IBH Pub

MAJOR IKS	Indian Knowledge Systems on Geography			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEO162M403

Learning objective*: The objective of this course is to enable students to examine the chronological development of Geography through the contributions of various scholars, approaches, and schools of thought, while understanding the major themes and components of Geography in relation to sustainable development, cultural heritage, environmental awareness, and human well-being aligned with relevant Sustainable Development Goals (SDGs), particularly SDG 4 (Quality Education), SDG 11 (Sustainable Cities and Communities), SDG 13 (Climate Action), and SDG 15 (Life on Land).*

Learning Outcomes:

After successful completion of the course the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO 1.	**Define** the various contributions of scholars in the field of Geography in India and relate them to knowledge development and quality education (SDG 4).	**BT 1**
CO 2.	**Interpret** the significance of geographical consciousness in India in relation to sustainable resource use, environmental awareness, and community development (SDG 11 & SDG 13).	**BT 2**
CO 3.	**Identify** the various Indian traditions associated with earth elements and evaluate their relevance to ecological conservation and sustainable living practices (SDG 13 & SDG 15).	**BT 3**
C0 4.	**Discover** the physical and humanistic perspectives and their dimensions in Geography in relation to physical and cultural surroundings, sustainability, and human-environment interaction (SDG 11 & SDG 15).	**BT 4**
CO 5.	**Explain** the significance of Indian culture in preserving traditional knowledge, cultural heritage, and social harmony for sustainable communities (SDG 4 & SDG 11).	**BT 5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Indian Geographical concepts: the earth, eclipses, latitudes and longitudes, cardinal points; weather and climate, Describe the earliest known Indian scholars and their knowledge on Earth and Gravity: Aryabhata and Bhaskara	14
Unit 2	Concepts of Panchatatva: Prithvi (Earth), Pavan (air), Jal (water), Tej (Solar energy) and Nabh (sky), Traditional practices of environmental conservation in India: Traditional water management practices, Indian Culture and conservation of biodiversity case studies from North East India (sacred forest of Mawphlang and Balpakram	10
Unit 3	Traditional housing designs of tribes, food habits, clothes etc, Agriculture in ancient India, Use of metals in ancient India: iron, steel, copper etc. The First Urbanisation in India: Indus Valley Civilization	12

Unit 4	Indian cultural sites and their significance in Indian spirituality; the idea of tirthas and their geographical implications; ideas of ancient Indian trade with South east Asian region, folk tales and their relevance in present day geography, customary land laws of North East India	12
	Total	**48**

Text Books:

1. Eck, D. L., 2013: India: A Sacred Geography, Harmony.
2. Mahadevan B. and Bhat V. R., 2022: Introduction To Indian Knowledge System: Concepts And Applications, PHI Learning.
3. Adhikari, S., 1992: Geographical Thought, Chaitanya Pub. House, Allahabad.
4. Dikshit, R. D., 1997: Geographical Thoughts: A Contextual History of Ideas, Prentice Hall of India, New Delhi.
5. Hussain, M., 2022: Evolution of Geographical Thought, Rawat Publication, New Delhi.

Reference Books:

5. Singh, R. L., (ed), 1971: India: A Regional Geography, National Geographical Society of India, Varanasi.
6. Pletcher. K., 2010: The Geography of India: Sacred and Historic Places (Understanding India), Britannica Educational Pub

MINOR	ECONOMIC GEOGRAPHY L-T-P-C: 3-0-0-3 Credit Units: 4 Scheme of Evaluation: (T+P)	Subject Code: GEO162N401

Course Objectives: *This course focuses on the basic concepts of Economic Geography and the associated spatial patterns and processes of primary economic activities across the world, with special emphasis on sustainable economic development, employment generation, resource utilization, and regional growth in alignment with SDG 8: Decent Work and Economic Growth.*

Course Outcomes:

After successful completion of the course, the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Define** the concepts and ways in which geographical aspects organize economic space and influence employment and economic activities.	BT1
CO2	**Compare** different sectors of the economy and arrive at logical conclusions regarding the importance of each sector in sustainable economic growth and decent work opportunities.	BT2
CO3	**Identify** the principles and significance of Economic Geography in relation to productivity, resource management, and regional economic development.	BT3
CO4	**Discover** new insights into the relationship between economy and geography and analyze contemporary economic and spatial challenges affecting sustainable growth and livelihoods.	BT4

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Meaning and scope of Economic Geography; Approaches in Economic Geography; Concept and classification of economic activity; factors influencing economic activities: Land, Labour, Capital; and production	10
Unit 2	Primary Activities: Subsistence and Commercial agriculture, forestry, fishing and mining; Secondary Activities: Manufacturing (Cotton Textile, Iron and Steel), Concept of Manufacturing Regions; Special Economic Zones and Technology Parks; Tertiary Activities: Transport, Trade and Services.	14
Unit 3	Theories of Economic Geography: Von Thunen's model of agricultural location, Industrial location theories of Weber, and concept of economic development by Myrdal and Rostow.	14
Unit 4	Economic Geography of Resources; Global pattern of distribution and production of selected resources: Food grains, iron ore, coal, petroleum and atomic minerals; Global economic scenario.	10
	Total	**48**

Text Books:

3. Alexander J. W., 1963: ***Economic Geography***, Prentice-Hall Inc., Englewood Cliffs, New Jersey.
4. Coe N. M., Kelly P. F. and Yeung H. W., 2007: ***Economic Geography: A Contemporary Introduction***, Wiley-Blackwell.

Reference Books:

6. Wheeler J. O., 1998: *Economic Geography*, Wiley..
7. Durand L., 1961: *Economic Geography*, Crowell.
8. Bagchi-Sen S. and Smith H. L., 2006: *Economic Geography: Past, Present and Future*, Taylor and Francis.
9. Willington D. E., 2008: *Economic Geography*, Husband Press.
10. Clark, G. L., Feldman, M. P., Gertler, M. S., & Williams, K. (Eds.). (2003). *The Oxford handbook of economic geography*. Oxford University Press

Minor	REGIONAL DEVELOPMENT			Subject Code:
	L-T-P-C: 3-0-0-3	Credit Units: 4	Scheme of Evaluation: (T)	GEO162N402

Course Objectives: *This course intends to help students understand the concept of regional development from a geographical perspective and its implications in the planning process, with special emphasis on inclusive growth, poverty alleviation, balanced regional development, and social equity in alignment with SDG 1: No Poverty and SDG 10: Reduced Inequalities.*

Course Outcomes:

After successful completion of the course, the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Define** the basic concepts of regional development and their relevance to poverty reduction and balanced growth.	BT1
CO2	**Explain** the strategic importance and applicability of regional development in reducing socio-economic and spatial inequalities.	BT2
CO3	**Build** development plans for rural and urban regions focusing on inclusive development, equitable resource distribution, and improved quality of life.	BT3
CO4	**Apply** the knowledge of regional development in real-world situations to address issues of poverty, marginalization, and regional disparities.	BT4
CO5	**Interpret** various issues related to regional development from national and global perspectives in the context of sustainable development, social justice, and reduced inequalities.	BT5

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Concept of region, regionalization, regionalism, and regional development; Types of Region; Significance of regional development studies; Approaches to regional development studies	12
Unit 2	Approaches to regional development in India; Role of Panchayati Raj Institutions in Regional Development; Concept of multi-level planning and Development: Micro, Meso and Macro.	12
Unit 3	Identification of resource regions; Concept of Development: Growth versus development; Concept of sustainable development and balanced development, Case studies of successful regional development initiatives in India.	12
Unit 4	Disparity of Regional Development in India; Development indicators; Measuring level of development, Regional Development theories and models: Growth Pole Model of Perroux; Cumulative Causation Theory of Gunnar Myrdal.	12
	Total	**48**

Text Book:

1. Krishnamurthy, J. (2000). **Rural Development - Problems and Prospects**. Jaipur, India: Rawat Publs.
2. Singh, R.B. (1985): **Geography of Rural Development**. New Delhi, India: Inter India.
3.
4. Alden J. and R. Morgan, 1974: ***Regional Planning: A Comprehensive View***, Leonard Hills Books, U.K.
5. Chand, M. and Puri, V. K. 1993: ***Regional Planning in India***, Allied Publishers Limited, B/M Asraf, Ali Road, New Delhi-110002.
6. Chandna, . R. C., 2000: ***Regional Planning: A Comprehensive Text***, Kalyani Publishers, New Delhi .

References:

1. Dickinson , R. E: ***City, Region and Regionalism,***
2. Hall, P., 1975: ***Urban and Regional Planning***, David and Charlos, London.
3. Hilborst,J. G. M. (1971) : ***Regional Planning: A System Approach***, Notterdam University Press.
4. Mishra, R. P, 1992: ***Regional Planning: Concept, Techniques, Policies and Case Studies,*** Concept Publications, New Delhi.

Type of Course: AEC **(w.e.f. 2023-24) UG programmes**

Semester: 4th Course Code: CEN982A401

Course Title: CEN IV – Employability and Communication Total credits: 1

Course level: 200

L-T-P-C: 1-0-0-1

Scheme of Evaluation: Theory and Practical

Course Objectives: This course is designed to enhance employability and maximize the students' potential by introducing them to the principles that determine personal and professional success, thereby helping them acquire the skills needed to apply these principles in their lives and careers.

Course Outcomes: After the successful completion of the course, the students will be able to

SI No	Course Outcome	Blooms Taxonomy Level
CO 1	**Demonstrate** understanding the importance of verbal and non-verbal skills while delivering an effective presentation.	BT 2
CO 2	**Develop** professional documents to meet the objectives of the workplace	BT 3
CO 3	**Define** and identify different life skills and internet competencies required in personal and professional life.	BT 3

Detailed Syllabus		
Units	**Course Contents**	**Periods**
I	**Presentation Skills** Importance of presentation skills, Essential characteristics of a good presentation, Stages of a presentation, Visual aids in presentation, Effective delivery of a presentation	5
II	**Business Writing** Report writing: Importance of reports, Types of reports, Format of reports, Structure of formal reports Proposal writing: Importance of proposal, Types of proposal, structure of formal proposals Technical articles: Types and structure	5
	Preparing for jobs Employment Communication and its Importance, Knowing the four-step employment process, writing resumes, Guidelines for a good	

III	resume, Writing cover letters Interviews: Types of interview, what does a job interview assess, strategies of success at interviews, participating in group discussions.	**5**
IV	**iteracy and Life Skills** **Digital literacy:** Digital skills for the '21st century', College students and technology, information management using Webspace, Dropbox, directory, and folder renaming conventions. Social Media Technology and Safety, Web 2.0. **Life Skills**: Overview of Life Skills: Meaning and significance of life skills, Life skills identified by WHO: self-awareness, Empathy, Critical thinking, Creative thinking, Decision making, problem- solving, Effective communication, interpersonal relationship, coping with stress, coping with emotion. Application of life skills: opening and operating bank accounts, applying for pan, passport, online bill payments, ticket booking, gas booking.	**5**

Keywords: Employability, business writing, presentation skills, life skills

Text:

1. *Business Communication* by PD Chaturvedi and Mukesh Chaturvedi

References:

1. *Business Communication* by Shalini Verma References:

2. *Technical Communication* by Meenakshi Raman and Sangeeta Sharma

Credit Distribution		
Lecture/Tutorial	**Practicum**	**Experiential Learning**
15 hours	-	10 hours - Movie/ Documentary screening - Field visits - Peer teaching - Seminars - Library visits

Subject Name: Behavioural Sciences -IV
UG 4th semester
Course code: BHS982A402
Credit: 1

Course objectives: To increase one's ability to draw conclusions and develop inferences about attitudes and behaviour, when confronted with different situations that are common in modern organizations.

Course outcomes: On completion of the course the students will be able to:

CO1**:** Understand the importance of individual differences- BT 1
CO2: Develop a better understanding of self in relation to society and nation- BT2
CO3: Facilitation for a meaningful existence and adjustment in society- BT 3

Modules	Course Contents	Periods
I	**Managing Personal Effectiveness** Setting goals to maintain focus, Dimensions of personal effectiveness (self-disclosure, openness to feedback and perceptiveness), Integration of personal and organizational vision for effectiveness, A healthy balance of work and play, Defining Criticism: Types of Criticism, Destructive vs Constructive Criticism, Handling criticism and interruptions.	**4**
II	**Positive Personal Growth** Understanding & developing positive emotions, Positive approach towards future, Impact of positive thinking, Importance of discipline and hard work, Integrity and accountability, Importance of ethics in achieving personal growth.	**4**
III	**Handling Diversity** Defining Diversity, Affirmation Action and Managing Diversity, Increasing Diversity in Work Force, Barriers and Challenges in Managing Diversity.	**4**
IV	**Developing Negotiation Skills** Meaning and Negotiation approaches (Traditional and Contemporary) Process and strategies of negotiations. Negotiation and interpersonal communication. Rapport Building – NLP.	**4**
Total		**16**

Text books:

- J William Pfeiffer (ed.) Theories and Models in Applied Behavioural Science, Vol 3, Management; Pfeiffer &Company
- Blair J. Kolasa, Introduction to Behavioural Science for Business, John Wiley & Sons Inc.

Semester 5

B.A./B. Sc. (Honours) Course in Geography: Semester-V

Paper I Major	REGIONAL PLANNING AND DEVELOPMENT L-T-P-C: 4-0-0-4 Credit Units: 4 (T) Scheme of Evaluation:	Subject Code: GEO162M501

Course Objectives: This course intends to make students understand the concept of a region from a Geographic perspective and its ramifications in planning. The course also aims to promote awareness of global development goals, particularly **SDG 1: No Poverty** (sub-themes: 1.1 Poverty eradication, 1.4 Access to resources) and **SDG 9: Industry, Innovation & Infrastructure** (sub-themes: 9.1 Infrastructure, 9.5 Innovation), by enabling students to critically engage with issues of regional disparity, equitable resource distribution, and the role of infrastructure in driving inclusive regional development.

Course Outcomes:

After successful completion of the course, the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Define** basic concepts of regional planning and relate them to frameworks of poverty eradication and access to resources (SDG 1.1, 1.4).	BT1
CO2	**Explain** the strategic importance of multi-level planning and the role of infrastructure in regional development with reference to SDG 9.1.	BT2
CO3	**Build** plans for development in rural and urban regions that integrate inclusive resource access and innovation-driven strategies aligned with SDG 9.5.	BT3
CO4	**Apply** this knowledge to real-world situations, linking regional planning outcomes to poverty reduction goals of SDG 1.	BT4
CO5	**Interpret** various issues related to regional planning on national and global perspectives through the lens of SDG 1 and SDG 9.	BT5

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Concept of region, regionalization, regionalism and regional development; Types of Regions; Meaning and purpose of regional planning; Approaches to regional planning	12
Unit 2	Identification of resource regions; Concept of Development: Growth versus development; Concept of sustainable development and balanced development, Case studies of regional planning exercises: National Capital Region and River basin planning- a case study from India	12

Unit 3	Decentralization and Multi-level planning - features of decentralised planning, decentralised planning in India, concept and procedures in multilevel planning; stages in the evolution of multi-level planning process, multi-level planning in India, Regional planning strategy under Five Year Plans, Regional Planning in India: Macro, meso and micro level planning; Local level planning and Panchayati Raj (GPDP); Participatory approach in planning; NITI Aayog	12
Unit 4	Disparity of Regional Development in India: Development indicators; Measuring level of development, Regional Development theories and models: Concept and basic ideas of Growth Pole Model of Perroux, Theory of Prebisch, Cumulative Causation Theory of Gunnar Myrdal, Stages of Economic Growth model of Rostow	12
	Total	**48**

Text Book:

7. Alden J. and R. Morgan, 1974: ***Regional Planning: A Comprehensive View***, Leonard Hills Books, U.K.
8. Bhat, L. S., 1976: ***Micro-Level Planning: A Case Study of Karnal Area, Haryana***, Concept Publishing Co., New Delhi.
9. Chand, M. and Puri, V. K. 1993: ***Regional Planning in India***, Allied Publishers Limited, B/M Asraf, Ali Road, New Delhi-110002.
10. Chandna, . R. C., 2000: ***Regional Planning: A Comprehensive Text***, Kalyani Publishers, New Delhi .

References:

5. Dickinson , R. E: ***City, Region and Regionalism,***
6. Hall, P., 1975: ***Urban and Regional Planning***, David and Charlos, London.
7. Hilborst,J. G. M. (1971) : ***Regional Planning: A System Approach***, Notterdam University Press.
8. Mishra, R. P, 1992: ***Regional Planning: Concept, Techniques, Policies and Case Studies,*** Concept Publications, New Delhi.

Major Paper II	POPULATION AND SETTLEMENT GEOGRAPHY L-T-P-C: 4-0-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEO162M 502

Course Objectives: This course aims to provide students with a comprehensive understanding of population dynamics, demographic processes, and patterns of human settlement. The course integrates the principles of **SDG 10: Reduced Inequalities** (sub-themes: 10.2 Social inclusion, 10.7 Migration) and **SDG 11: Sustainable Cities and Communities** (sub-themes: 11.1 Housing, 11.3 Urban planning), encouraging students to critically examine inequalities in population distribution, migration patterns, and the challenges of building inclusive, resilient, and sustainable human settlements.

Course Outcomes:

By the end of this course the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Define** key concepts of population geography and settlement patterns with reference to social inclusion as envisaged under SDG 10.2.	BT1
CO2	**Interpret** demographic transitions, migration trends, and their spatial implications in the context of SDG 10.7 (Migration).	BT2
CO3	**Identify** disparities in population distribution and settlement structures in relation to housing challenges outlined in SDG 11.1.	BT3
CO4	**Analyse** population-settlement relationships and urban planning issues in alignment with SDG 11.3 (Urban planning) for sustainable community development.	BT4
CO5	**Determine** clear exposition of spatial and structural characteristics of human settlements	BT5

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Field of Population Geography, its emergence as a branch of Geography, significance and relation with demography; Key concepts of Population pressure: under population, optimum population, over population; Components of population change: fertility, mortality and migration; and associated factors	12

Unit 2	Population Growth and Distribution: World and India, factors influencing population distribution; Measures of population density and distribution; Demographic transition model and theory of population growth by Malthus; concept of population resource relationship and population resource regions	12
Unit 3	Defining the field and scope of Settlement Geography; origin of settlements; rural and urban settlements: types of rural settlement, classification of urban settlement , functional classification of towns, law of primate city and rank size rule.	12
Unit 4	Concept of urbanization, rural-urban fringe, city region, settlement hierarchy with respect to central place theory (Christaller and Losch)	12
	Total	**48**

Text Books Suggested:

1. Singh R.Y. (Rep. 2010) - Geography of Settlements, Sharda Pustak Bhawan, Allahabad
2. Chandna R. C. (Rep.2010) – ***A Geography of Population, Concepts, Determinants and Patterns***, Kalyani Publishers, New Delhi.
3. Maurya S.D (Rep. 2018): ***Settlement Geography***, Sharda Pustak Bhawan, Allahabad
4. Sandram, K. V. and Nangia, S., (eds): ***Population Geography***, Heritage Publishers, New Delhi. Inc., New York.

Reference books:

1. Clarke, J. I., 1972**:** ***Population Geography*****,** Pergamon Press, Oxford.
2. Peters, G. L. and Larkin, R. P., 1979: ***Population Geography: Problems, Concepts and Prospects,*** Kendall/ Hunt Iowa.
3. Trewartha, G. T., 1969: ***A Geography of Population: World Pattern***, John Wiley & Sons.
4. Woods, R., 1979**:** ***Population Analysis in Geography***, Longman, London.
5. Robinson, H., 1981: ***Population and Resources***, Macmillan Press, London

Paper III MAJOR	DISASTER MANAGEMENT L-T-P-C: 4-0-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEO162M503

Course Objectives: This course is designed to equip students with the knowledge and skills required to understand the nature, causes, and management of natural and human-induced disasters. The course integrates the objectives of **SDG 3: Good Health and Well-being** (sub-theme: 3.9 Environmental health), **SDG 6: Clean Water and Sanitation** (sub-theme: 6.1 Safe water), and **SDG 13: Climate Action** (sub-theme: 13.1 Climate resilience), enabling students to appreciate the interdependence of disaster preparedness, environmental health, water security, and climate-resilient community planning.

Course Outcomes:

After successful completion of the course, the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	**Define** the concept of disaster, its types, and their linkages to environmental health concerns as outlined in SDG 3.9.	**BT 1**
CO2.	**Explain** the impacts of disasters on water safety and sanitation infrastructure with reference to SDG 6.1 (Safe water).	**BT 2**
CO3.	**Identify** disaster-prone regions and assess vulnerability using climate resilience frameworks consistent with SDG 13.1.	**BT 2**
CO4.	**Evaluate** disaster mitigation and preparedness strategies that simultaneously promote environmental health, safe water access, and climate-resilient communities.	**BT 4**
CO5.	**Recommend** integrated disaster management plans that address the goals of SDG 3, SDG 6, and SDG 13 at local, national, and global levels.	**BT 5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Disasters: Definition and Concepts: Hazards, Disasters; Risk and Vulnerability; Classification	12
Unit 2	Disasters in India: Flood-Causes, Impact, Distribution and Mapping; Landslide-Causes, Impact, Distribution and Mapping; Drought- Causes, Impact, Distribution and Spatial Pattern	12
Unit 3	Disasters in India: Earthquake and Tsunami- Causes, Impact, Distribution and Mapping; Cyclone- Causes, Impact, Distribution and Spatial Pattern; Manmade disasters- Causes, Impact, Distribution and Spatial Pattern	12

Unit 4	Response and Mitigation to Disasters: Mitigation and Preparedness, NDMA and NIDM; Indigenous Knowledge and Community-Based Disaster Management; Do's and Don'ts During and Post Disasters	12
	Total	**48**

Text Book:

1. Singh, R. B. (ed.), (2006) Natural Hazards and Disaster Management: Vulnerability and Mitigation, Rawat Publications, New Delhi.
2. Sinha, A. (2001). Disaster Management: Lessons Drawn and Strategies for Future, New United Press, New Delhi.

References:

1. Government of India. (1997) Vulnerability Atlas of India. New Delhi, Building Materials & Technology Promotion Council, Ministry of Urban Development, Government of India.
2. Kapur, A. (2010) Vulnerable India: A Geographical Study of Disasters, Sage Publication, New Delhi.
3. Modh, S. (2010) Managing Natural Disaster: Hydrological, Marine and Geological Disasters, Macmillan, Delhi.
4. Singh, R.B. (2005) Risk Assessment and Vulnerability Analysis, IGNOU, New Delhi.
5. Stoltman, J.P. et al. (2004) International Perspectives on Natural Disasters, Kluwer Academic Publications. Dordrecht.
6. Singh Jagbir (2007) "Disaster Management Future Challenges and Opportunities", 2007. Publisher- I.K. International Pvt. Ltd. S-25, Green Park Extension, Uphaar Cinema

Minor	POPULATION AND POLITICAL GEOGRAPHY L-T-P-C: 4-0-0-4 (T) Credit Units: 4 Scheme of Evaluation:	Subject Code: GEO162N501

Course Objectives: This course aims to introduce students to the theoretical and applied dimensions of population geography and political geography, focusing on spatial patterns of population and the political organization of space. The course promotes awareness of **SDG 10: Reduced Inequalities** (sub-theme: 10.2 Social inclusion) and **SDG 16: Peace, Justice and Strong Institutions** (sub-themes: 16.6 Governance, 16.7 Inclusive institutions), encouraging students to critically assess how demographic dynamics and geopolitical processes intersect with issues of equity, governance, and inclusive political participation.

Course Outcomes:

By the end of this course the students will be able to:		
CO.	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Define** the parameters of population geography and political geography and relate them to principles of social inclusion (SDG 10.2).	BT1
CO2	**Interpret** spatial patterns of population and political boundaries in the context of equitable governance (SDG 16.6).	BT2
CO3	**Identify** political geographic concepts such as state, territory, and boundaries in relation to inclusive institutions and democratic participation (SDG 16.7).	BT3
CO4	**Analyse** how population dynamics and political processes reinforce or reduce inequalities across regions, aligned with SDG 10.2 and SDG 16.	BT4
CO5	**Appraise** the socio-cultural and political theme in the geographical dimensions	BT5

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Significance of Population and relation with demography; Key concepts of Population pressure: under population, optimum population, over population; Components of population change: fertility, mortality and migration; and associated factors	12
Unit 2	Population Growth and Distribution: World and India, factors influencing population distribution; Measures of population density and distribution; Demographic transition model and theory of population growth by Malthus; concept of population resource relationship and population resource regions	12
Unit 3	Nature, scope and subject matter of political geography; Approaches to the study of political geography, The field and school of thoughts in political geography: landscape school, ecology school	12
Unit 4	Concepts in political geography: frontier and boundary (with reference to India), lebensraum, state and nation, core-periphery and capital, buffer zone, federal state, Colonialism, desalinization, theories of Rim-land and Heartland.	12
	Total	**48**

Text Books Suggested:

5. Singh R.Y. (Rep. 2010) - Geography of Settlements, Sharda Pustak Bhawan, Allahabad
6. Chandna R. C. (Rep.2010) – ***A Geography of Population, Concepts, Determinants and Patterns***, Kalyani Publishers, New Delhi.
7. Maurya S.D (Rep. 2018): ***Settlement Geography***, Sharda Pustak Bhawan, Allahabad
8. Sandram, K. V. and Nangia, S., (eds): ***Population Geography***, Heritage Publishers, New Delhi. Inc., New York.

Reference books:

6. Clarke, J. I., 1972**:** ***Population Geography*****,** Pergamon Press, Oxford.
7. Peters, G. L. and Larkin, R. P., 1979: ***Population Geography: Problems, Concepts and Prospects,*** Kendall/ Hunt Iowa.
8. Trewartha, G. T., 1969: ***A Geography of Population: World Pattern***, John Wiley & Sons.
9. Woods, R., 1979**:** ***Population Analysis in Geography***, Longman, London.
10. Robinson, H., 1981: ***Population and Resources***, Macmillan Press, London

	Internship Credit Units: 4 Scheme of Evaluation: (P)	GEO162M5 24

Course Objectives: *This paper provides an understanding the basics of research project preparation.*

Course Outcomes:

After the completion of course, the students will have ability to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Relate** real world issues for carrying out research on a specific field	BT1
CO2	**Infer** ideas of research through literature review.	BT2
CO3	**Develop** hypotheses and research questions.	BT3
CO4	**Identify** appropriate sampling techniques.	BT4
CO5	**Interpret** the various types of data along with **critical evaluation**.	BT5
CO6	**Design and develop** a detailed project report	BT6

Detailed Syllabus:

Modules	**Topics and Course Content**	**Periods**
Unit 1	Internship in institutes, organizations, and firms / industry of repute in Northeast India.	24
Unit 2	Preparation of internship report in prescribed format during 4 week or 120 hours of the commencement of course of 5th semester. Submission of the report after a week of the announcement of routine for 5th End Semester Examination duly signed by the appropriate person in the institute concerned.	24
	Total	**48**

Note: Students will work as an intern during the semester break for 4 weeks after 4th semester. Students not being able to obtain any internship will be assigned project work from the department.

Text Books:

As per the list of given in syllabus based on topic selected

Reference Books:

As per the list of given in syllabus based on topic selected

Learning Outcomes:

After the completion of course, the students will have ability to:

- Prepare and execute major project by collecting primary and/or secondary data
- Improve the skill of organizing the study based on project / research objectives
- Process, analyse the data and write scientific project report

Semester *6*

SEMESTER 6

Paper I Core Course	GEOGRAPHY OF INDIA L-T-P-C: 4-0-0-4	Credit Units: 4 (T)	Scheme of Evaluation:	Subject Code: GEO162M601

Course Objectives: This course provides a systematic and regional understanding of India's physical, agricultural, and economic geography. The course is oriented towards global development goals, particularly **SDG 2: Zero Hunger** (sub-theme: 2.1 Food security) and **SDG 7: Affordable and Clean Energy** (sub-themes: 7.1 Energy access, 7.2 Renewable energy), by enabling students to examine India's agricultural systems, food production challenges, and the spatial dimensions of energy infrastructure and the transition to renewable energy sources.

Course Outcomes:

By the end of this course the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Define** the physical and agricultural framework of India and relate it to food security objectives of SDG 2.1.	BT1
CO2	**Explain** the spatial distribution of India's energy resources and their relevance to affordable energy access as per SDG 7.1.	BT2
CO3	**Identify** regions of agricultural strength and vulnerability in India in the context of achieving Zero Hunger (SDG 2).	BT3
CO4	**Analyse** the geographic dimensions of India's transition to renewable energy and its implications for SDG 7.2.	BT4
CO5	**Examine** the position of India in global context.	BT5

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	India as a geographical entity, Location and situation; India in the context of neighbouring counties, Physical background of regional development: relief, drainage system, climate, soil and natural vegetation; Indian monsoon: mechanism and characteristics.	10
Unit 2	Population and development issues: population growth and its socio-economic implications, age-sex composition, literacy, urbanization, occupation and social composition (tribes, caste and religion).	14
Unit 3	Green Revolution, Cropping pattern, Agro-climatic regions, Mineral and power resources distribution and development (iron ore, coal, petroleum and hydro- electric power and solar energy); Industrial regions of India.	12

Unit 4	Regional disparities in economic development: Agriculture, industry and transport and communication, India's geo-economic position in Asia and the world; its economic development policies and international relations.	12
	Total	**48**

Text Books:

7. Singh, R. L., (ed), 1971: India: A Regional Geography, National Geographical Society of
8. India, Varanasi.
9. Bhatt, L. S., 1973: Regional Planning in India, Statistical Publishing Society, Calcutta.
10. Tirtha R. & Gopal Krishna, 1996: Emerging India Reprinted by Rawat Publications, Jaipur.

Reference Books:

5. Dreze, Jean & Amartya Sen (ed.), 1996: India Economic Development and Social opportunity, Oxford University Press, New Delhi.
6. Kundu A. Raza Moonis, 1982: Indian Economy: the Regional Dimension. Spectrum
7. Publishers, New Delhi.

Paper I Core Course	GEOGRAPHICAL THOUGHT L-T-P-C: 4-0-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEO162M602

Course objective: This course traces the historical evolution of geographical ideas, philosophies, and paradigms from ancient to contemporary times. The course integrates the principles of **SDG 4: Quality Education** (sub-theme: 4.7 Education for sustainable development) and **SDG 5: Gender Equality** (sub-theme: 5.1 Gender equality), by encouraging students to critically examine how geographical knowledge has evolved, how feminist geographies have challenged dominant narratives, and how education in geographical thought can foster sustainability-oriented and gender-sensitive perspectives.

Course Outcomes:

By the end of this course the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Define** the various contributions of scholars and schools of thought in the development of Geography, linking classical knowledge to education for sustainable development (SDG 4.7).	BT1
CO2	**Interpret** the paradigm shifts in geographical thought, including the emergence of feminist geography and gender-sensitive approaches consistent with SDG 5.1.	BT2
CO3	**Identify** the relationship between geographical thought and the evolution of sustainability discourse in the context of SDG 4.7.	BT3
CO4	**Discover** how theoretical perspectives in geography reflect and address gender inequalities, aligning with the ideals of SDG 5.1.	BT4
CO5	**Explain** the various issues of real world with a geographical perspective	BT5

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Place of geography in the classification of knowledge: Defining the field of geography, relation of geography with other natural and social sciences; Defining the field of human geography; nomothetic and ideographic approaches	12
Unit 2	Geography through the ages; general character of geographic knowledge during the ancient and mediaeval period; impact of discoveries and European renaissance on the emergence of modern geography, Foundations of modern geography: Contribution of German (Humboldt, Ritter, Ratzel), French (Paul Vidal de la Blache), British and American geographers.	12
Unit 3	Evolution of geographic thought (Determinism with special reference to Darwinism, Neo-Determinism, Possibilism, Geography as Human Ecology, Geography as the study of areal differentiation, Morphology of Landscape.)	10

Unit 4	Positivism and quantitative revolution, spatial organization, locational analysis, behaviouralism, humanism, radicalism; gender geography and post modernism geography	14
	Total	**48**

Text Books:

6. Adhikari, S., 1992: Geographical Thought, Chaitanya Pub. House, Allahabad.
7. Berry, B. J. L., 1973: 'A Paradigm for Modern Geography', in R. J. Chorley (ed), Directions in Geography, London Methuen.
8. Bunge, W., 1962: Theoretical Geography, Lund Studies in Geography, Lund, C.W.K. Gleerup.
9. Buttimar, A., 1978: 'On People, Paradigms and Progress in Geography', in D.R. Stoddart (ed),
10. Geography, Ideology and Social Concern, Oxford, Blackwell.
11. Dickinson, R. E., 1969: Makers of Modern Geography, Routledge and Kegan Paul, London.

Reference Books:

1. Dikshit, R. D., 1997: Geographical Thoughts: A Contextual History of Ideas, Prentice Hall of India, New Delhi.
2. Gold, J. R., 1980: An Introduction to Behavioural Geography, Oxford University Press.
3. Hartshorne, R., 1939: The Nature of Geography, Association of American Geographers, Lancaster, Penn.
4. Hartshorne, R., 1959: Perspective on the Nature of Geography, Rand Mckully, Chicago.
5. Harvey, D., 1969: Explanation in Geography, St. Martin's Press, New York.

Paper MAJOR	AGRICULTURAL GEOGRAPHY			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEO162M603

Course Objectives: This course aims to provide a comprehensive understanding of the spatial patterns of agriculture, agricultural systems, and agrarian development at regional and global scales. The course is anchored in **SDG 2: Zero Hunger** (sub-themes: 2.3 Agricultural productivity, 2.4 Sustainable agriculture), **SDG 6: Clean Water and Sanitation** (sub-theme: 6.4 Water-use efficiency), and **SDG 15: Life on Land** (sub-theme: 15.1 Forest conservation), equipping students to evaluate the sustainability of agricultural practices with respect to food production, water resource management, and the conservation of terrestrial ecosystems.

Course Outcomes:

After successful completion of the course, the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	Define the scope and significance of agricultural geography and relate global agricultural systems to food security and productivity (SDG 2.3).	**BT 1**
CO2.	Explain the principles of sustainable agriculture and their role in ensuring food security in the context of SDG 2.4.	**BT 2**
CO3.	Identify the linkages between agricultural water demand, irrigation systems, and efficient water-use practices aligned with SDG 6.4.	**BT 3**
CO4.	Analyse the impact of agricultural expansion on land degradation and forest ecosystems in the context of SDG 15.1.	**BT 4**
CO5.	Evaluate integrated land and water management strategies in agriculture that simultaneously address SDG 2, SDG 6, and SDG 15 goals.	**BT 5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Trends in the study of agricultural geography, Approaches to the study of agricultural geography: commodity, systematic, regional, inductive and deductive, Determinants of agriculture: physical, economic, social, institutional and technological, concept of Land holding and land tenure systems; Land reforms, land use policy and planning	12
Unit 2	Factors of agricultural production: land, labour, capital, organization and govt. policy, Concept of large-scale and small-scale agriculture , Concept of agricultural region, agricultural types and agricultural systems , Methods in agricultural geography: Von Thunen's model of agricultural location and Franklin's type of farming model	12

Unit 3	Methods of agricultural regionalization: crop-combination, crop concentration and diversification, crop intensity, degree of commercialisation , crop rotation and agricultural efficiency, Agricultural productivity: concept and measurement, factors influencing crop productivity, Cropping pattern, factors influencing cropping pattern, shifting in cropping pattern, crop and cropping hazards, crop and crop land management	14
Unit 4	Agricultural classification: according to Whittlesey, Richard Thoman and Peter Corbin, Land use and land capability classification, Agricultural development experiences of USA and Israel.	10
	Total	**48**

Text Books:

1. Anderson, J.R., 1970: A Geography of Agriculture, Iowa: WMC Brown Co. Clark, Colin and Haswell, Margaret, 1964: The Economy of Subsistence Agriculture, St. Martin's, London.
2. Chorley, R. J. and Haggett, P., 1971: Socio-Economic Models in Geography, Methuen and Co. Ltd., London.
3. Dunn, E. S., 1954: The Location of Agricultural Production, University of Florida Press, Gainsville.

Reference Books:

1. Hussain, M., 2001: Systematic Agricultural Geography, Rawat Publication, Jaipur and New Delhi.
2. Morgan, W.B. and Munton, R.J.C., 1971: Agricultural Geography, Methuen, London.
3. Singh, J., 1974: Agricultural Atlas of India: A Geographical Analysis, Vishal Publishers, Kurukhsetra.
4. Singh, J., 1976: Agricultural Geography, Tata McGraw Hill Pub. Co., New Delhi.
5. Symons, L., 1967: Agricultural Geography, G. Bells and Sons, London.

Learning Outcomes:

By the end of this course the students will be able to

- Understand the principles of agricultural geography
- Understand various concepts related to agricultural geography
- Understand the strategic importance and applicability of agricultural geography in national and global aspect

Paper MAJOR	CARTOGRAPHIC TECHNIQUES GEOGRAPHY L-T-P-C: 0-0-8-4 Credit Units: 4 Scheme of Evaluation: (P)	Subject Code: GEO162M614

Course Objectives: This course is designed to develop students' competence in the principles and techniques of cartography, map design, and spatial data visualization. The course aligns with **SDG 9: Industry, Innovation and Infrastructure** (sub-theme: 9.5 Innovation), promoting the use of innovative cartographic tools and geospatial technologies to solve real-world geographic problems and contribute to evidence-based planning and decision-making.

Course Outcomes:

By the end of this course the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Define** the fundamental principles of cartography and map projections as tools for innovation in spatial communication (SDG 9.5).	BT1
CO2	**Explain** the design and construction of various types of maps and their relevance to technological innovation in geographic information (SDG 9.5).	BT2
CO3	**Apply** appropriate cartographic techniques to represent geographic data, contributing to innovation-driven geographic analysis.	BT3
CO4	**Evaluate** the effectiveness of cartographic representations in addressing geographic, developmental, and environmental challenges consistent with SDG 9 goals.	BT4

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Drainage basin delimitation , Ordering of streams, calculation of bifurcation ratio, length ratio, drainage frequency and density, calculation of area by planimeter; Preparation of rainfall variability map (Assam and Rajasthan), drawing of hythergraph, climograph and ergograph	10
Unit 2	Map Projection: Basic concepts, classification, basic Principles of construction of zenithal, conical and cylindrical groups of map projections Construction of graticules and drawing of maps thereon including properties and uses of: Zenithal Gnomonic Projection (Equatorial case) Sinusoidal Projection	10

	Conical Projection with one-standard parallels Lambert's Conical Equal Area Projection Mercator's Projection	
Unit 3	Geodetic and plane table surveying; Principles and techniques of surveying by Plane Table, Prismatic Compass	10
Unit 4	Field techniques: levelling by Dumpy Level and Transit Theodolite (Base accessible and inaccessible method)	18
	Total	**24**

Reference Books:

1. Weiesner, C. J.: ***Hydrometeorology***, Chapman & Hall Ltd.
2. Gregory, K, J. and Walling, D.E., 1973: Drainage Basin- Form and Process, Edward Arnold, London
3. Goudie, Andrew, et. Al. (eds), 1981: Geomorphological Techniques, George Allen & Unwin, London.
4. Woods, R., 1979**:** ***Population Analysis in Geography***, Longman, London.

Minor	GEOGRAPHY OF DEVELOPMENT OF INDIA L-T-P-C: 4-0-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEO162N601

Course Objectives: This course examines the geographic dimensions of development in India, including regional disparities, resource distribution, and economic growth patterns. It is aligned with **SDG 1: No Poverty** (sub-theme: 1.1 Poverty eradication) and **SDG 8: Decent Work and Economic Growth** (sub-theme: 8.3 Employment), enabling students to critically assess India's development trajectories, persistent poverty pockets, and the role of employment generation and inclusive economic growth in achieving sustainable development.

Course Outcomes:

By the end of this course, the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	**Define** the concept of development and relate regional disparities in India to poverty eradication goals of SDG 1.1.	**BT 1**
CO2.	**Explain** the significance of employment generation and decent work opportunities in driving inclusive development in India (SDG 8.3).	**BT 2**
CO3.	**Identify** indicators of development and underdevelopment across Indian regions in relation to SDG 1 and SDG 8 frameworks.	**BT 3**
CO4.	**Analyse** how policy interventions in India address the dual imperatives of poverty reduction (SDG 1.1) and economic empowerment (SDG 8.3).	**BT 4**
CO5.	**Examine** the position of India in global context.	**BT 5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	India as a geographical entity, Location and situation; India in the context of neighbouring counties, Physical background of regional development: relief, drainage system, climate, soil and natural vegetation, major crops, Indian monsoon: mechanism and characteristics, natural disaster in India (earthquake, drought, flood, cyclone, tsunami, Himalayan highland hazards)	10

Unit 2	Population and development issues: population growth and its socio-economic implications, literacy, urbanization, occupation and social structure and development inequalities, tribes and religion	14
Unit 3	Mineral and power resources and development: iron ore, coal, petroleum and hydro- electric power potential, agro-climatic and physiographic divisions of India, agro climatic regions, industrial regions of India	12
Unit 4	Regional disparities in economic development: Agriculture, industry and transport and communication, India's geo-economic position in Asia and the world; its economic development policies and international relations.	12
	Total	**48**

Text Books:

11. Singh, R. L., (ed), 1971: India: A Regional Geography, National Geographical Society of
12. India, Varanasi.
13. Bhatt, L. S., 1973: Regional Planning in India, Statistical Publishing Society, Calcutta.
14. Tirtha R. & Gopal Krishna, 1996: Emerging India Reprinted by Rawat Publications, Jaipur.

Reference Books:

8. Dreze, Jean & Amartya Sen (ed.), 1996: India Economic Development and Social opportunity, Oxford University Press, New Delhi.
9. Kundu A. Raza Moonis, 1982: Indian Economy: the Regional Dimension. Spectrum
10. Publishers, New Delhi.
11. Robinson, Francis, 1989 : The Cambridge Encyclopaedia of India, Pakistan, Bangladesh, Sri Lanka, Nepal, Bhutan & Maldives. Cambridge University Press, London.

Semester 7

Major	GEOGRAPHY OF RURAL DEVELOPMENT	Subject Code:
	L-T-P-C: 4-0-0-4 Credit Units: 4 Scheme of Evaluation: (T)	GEO162M701

Course Objectives: This course aims to provide students with a thorough understanding of the processes, policies, and patterns of rural development in India and across the world. It is grounded in **SDG 1: No Poverty** (sub-theme: 1.1 Poverty eradication), **SDG 2: Zero Hunger** (sub-theme: 2.4 Sustainable agriculture), and **SDG 10: Reduced Inequalities** (sub-theme: 10.2 Social inclusion), enabling students to evaluate rural development strategies that address poverty, promote sustainable agricultural livelihoods, and foster social inclusion in marginalized rural communities.

Course Outcomes:

After successful completion of the course, the students will be able to:

CO	Course Outcome	Blooms Taxonomy Level
CO1	**Define** the concept of rural development and relate rural poverty to poverty eradication goals of SDG 1.1.	BT1
CO2	**Explain** the role of sustainable agriculture and agro-based livelihoods in rural development, consistent with SDG 2.4.	BT2
CO3	**Identify** disparities in rural social inclusion and welfare delivery mechanisms in the context of SDG 10.2.	BT3
CO4	**Analyse** the effectiveness of rural development programmes in India through the triple lens of poverty elimination, sustainable agriculture, and social inclusion (SDG 1, 2, 10).	BT4
CO5	**Recommend** integrated rural development strategies that simultaneously address food security, poverty, and inequality as per the SDG framework.	BT5
CO6	**Design and develop** a detail project report	BT6

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Rural Development: meaning and dimensions; Need for Rural Development studies in geography; Rural Poverty; Rural development in India: Role of Zilla Parishad and Panchayats in Rural Development, Rural Finances – Banks, NABARD etc.	10
Unit 2	Concept of Village, Rural Settlement pattern, Rural Economic Base: Panchayati raj System, Agriculture and Allied Sectors, Seasonality and Need for Expanding Non-Farm Activities, Co-operatives	14
Unit 3	The Problem of Housing, housing types, low cost houses, the housing schemes in rural area, Dimensions of Rural unemployment and under employment; Rural – Urban migration issues.	10

Unit 4	Recent Rural Development programmes in India: MGNREGA, DAY-NRLM, DDU-GKY, PMAY-G, SPMRM, PMKSY, PM-KUSUM, SVAMITVA Scheme, PM Surya Ghar: Muft Bijli Yojana, PM Vishwakarma Yojana	14
	Total	**48**

Text Book:

1. Gilg A. W., 1985: An Introduction to Rural Geography, Edwin Arnold, London.
2. Krishnamurthy, J. 2000: Rural Development - Problems and Prospects, Rawat Publs., Jaipur
3. Lee D. A. and Chaudhri D. P. (eds.), 1983: Rural Development and State, Methuen, London.
4. Misra R. P. and Sundaram, K. V. (eds.), 1979: Rural Area Development: Perspectives and Approaches, Sterling, New Delhi.

References:

1. Robb P. (ed.), 1983: Rural South Asia: Linkages, Change and Development, Curzon Press.
2. UNAPDI 1986:Local Level Planning and Rural Development: Alternative Strategies. (United Nations Asian & Pacific Development Institute, Bangkok), Concept Publs. Co., New Delhi.
3. Wanmali S., 1992: Rural Infrastructure Settlement Systems and Development of the Regional Economy in South India, International Food Policy Research Institute, Washington, D.C.
4. Yugandhar, B. N. and Mukherjee, Neela (eds.) 1991: Studies in Village India: Issues in Rural Development, Concept Publs. Co., New Delhi.
5. Misra, R. P. (ed.), 1985: Rural Development: Capitalist and Socialist Paths, Vol. 1, Concept, New Delhi

MAJOR	Urban Geography L-T-P-C: 4-0-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEO162M702

Course Objectives: This course provides a systematic study of urbanization, urban structure, urban planning, and the challenges of managing growing cities. The course is anchored in **SDG 11: Sustainable Cities and Communities** (sub-themes: 11.1 Housing, 11.3 Urban planning, 11.5 Disaster resilience), enabling students to critically examine issues of urban growth, housing inequity, urban land use planning, and the resilience of cities to natural and human-induced disasters.

Course outcomes:

After successful completion of the course, the students will be able to:		
CO	**Course Outcome**	**BT Level**
CO1	**Define** the concept of urbanization and relate urban housing challenges to the goals of SDG 11.1.	**BT 1**
CO2	**Explain** theories and models of urban structure and their implications for inclusive and sustainable urban planning in the context of SDG 11.3.	**BT 2**
CO3	**Identify** urban vulnerabilities and the role of disaster-resilient planning strategies consistent with SDG 11.5.	**BT 3**
CO4	**Analyse** contemporary urban challenges — housing deficits, sprawl, transport, and disaster risk — through the framework of SDG 11.	**BT 4**
CO5	**Evaluate** urban development policies and master plans for their alignment with sustainable, inclusive, and disaster-resilient community goals (SDG 11.1, 11.3, 11.5).	**BT 5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Urban Geography: Meaning, subject matter and scope; approaches and trends in urban geography; Towns: Types, characteristics, origin and growth in global and national contexts; Functional classification of towns; Schemes of city classification	10
Unit 2	Patterns of Urbanization in developed and developing countries; Components of urbanization and urban population growth; Urban morphology and land use structure; Theories on the internal structure of town: concentric zone theory of Burgess and the Sector Theory of Hoyt	14
Unit 3	Concept of city-region, urban agglomeration, urban sprawl, Umland and periphery, rural-urban dichotomy and continuum, urban fringe, satellite town, new town, smart city; Urban Systems: Concept of urban system and hierarchy; Christaller's Central Place Theory; the	14

	rank-size distribution of cities; concept of primate city.	
Unit 4	Urban issues and problems: Housing, slums, civic amenities (transportation and drinking water), traffic congestion, pollution (air, noise, water), and crime. Urbanization and urban development planning in India: Trend and regional patterns of urbanization; national urban development policies and programmes; emerging urban issues of selected cities (Delhi NCR, Mumbai, Guwahati).	10
	Total	**48**

Text Book:

1. Bansal, S.C. (2010): Urban Geography, Meenakshi Prakashan, Meerut.
2. Hall T., 2006: Urban Geography, Taylor and Francis.
3. Kaplan D. H., Wheeler J. O. and Holloway S. R., 2008: Urban Geography, John Wiley.
4. Knox P. L. and McCarthy L., 2005: Urbanization: An Introduction to Urban Geography, Pearson Prentice Hall New York.
5. Pacione M., 2009: Urban Geography: A Global Perspective, Taylor and Francis.

References:

1. Bala, R. (1986): Urbanisation in India, Rawat, Jaipur.
2. Fyfe N. R. and Kenny J. T., 2005: The Urban Geography Reader, Routledge.
3. Graham S. and Marvin S., 2001: Splintering Urbanism: Networked Infrastructures, Technological Mobilities and the Urban Condition, Routledge

MAJOR III	GEOGRAPHY OF NORTHEAST INDIA AND ASSAM L-T-P-C: 4-0-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEO162M703

Course Objectives: This course offers a comprehensive regional geographic study of North East India and Assam, covering its physical setting, natural resources, cultural diversity, and developmental challenges. It is aligned with **SDG 15: Life on Land** (sub-themes: 15.1 Forest conservation, 15.5 Biodiversity protection), enabling students to understand the ecological richness of the region, the significance of its forests and biodiversity hotspots, and the challenges of balancing development with the conservation of terrestrial ecosystems.

Course Outcomes:

By the end of this course the students will be able to:		
CO.	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Define** the physical and cultural geographic character of North East India and Assam, highlighting the region's ecological significance in the context of SDG 15.	BT1
CO2	**Explain** the role of forests and biodiversity in sustaining the livelihoods and ecological balance of the region, consistent with SDG 15.1 (Forest conservation).	BT2
CO3	**Identify** the biodiversity hotspots of North East India and the threats to endemic species, linking findings to SDG 15.5 (Biodiversity protection).	BT3
CO4	**Analyse** the conflicts between developmental pressures and forest/biodiversity conservation in the context of SDG 15 goals for North East India and Assam.	BT4

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	North East India: location and strategic significance; Physical characteristics and their relation to development: Relief, drainage, climate, soil and vegetation and bio-diversity, Position of North-East India in relation to India and its neighbours, Socio-economic and trade relation with ASEAN countries	12
Unit 2	Population and Development: Population growth, composition and distribution, migration, population characteristics, social structure: race, caste, religious and linguistic composition.	12
Unit 3	Natural resources, their utilization and development: Coal, petroleum, natural gas, water and forests in North East India; Agriculture and Development: Agricultural modernization and strategies for future development; constraints of Industrial development; problems and prospects of tourism.	12

Unit 4	Locational significance of Assam; Position of Assam in relation to rest of India, its physical characteristics and their relation to development: Relief, drainage, climate, soil and vegetation, Population growth, composition and distribution in North-East India and Assam; Characteristics and status of agriculture, industry and tourism in Assam	12
	Total	**48**

Text Books:

1. Dutta Ray, B., et. al (eds), 2000: Population, Poverty and Environment in North East India,
2. Concept Publishing Co., New Delhi.
3. Taher M. and Ahmed, P., 2000: Geography of North East India, Mani-Manik Prakash, Guwahati.
4. Bhagabati, A. K. et al, 2001: Geography of Assam, Rajesh Publications, New Delhi.

Reference Books:

1. Barua, P. C., 1990: Development Planning of North East India, Mittal Publications, New Delhi.

North East India Geographical Society: North Eastern Geographer, Department of Geography, Gauhati University.

MAJOR IV	FIELD TECHNIQUES IN GEOGRAPHY L-T-P-C: 0-0-8-4 Credit Units: 4 (P) Scheme of Evaluation:	Subject Code: GEO162M714

Course Objectives: This course equips students with practical skills in field data collection, observation, survey techniques, and report preparation essential for geographic research and applied work. The course aligns with **SDG 4: Quality Education** (sub-theme: 4.4 Skills for employment) and **SDG 9: Industry, Innovation and Infrastructure** (sub-theme: 9.5 Innovation), fostering hands-on competencies that enhance students' employability and their capacity to apply innovative geospatial and field methods to address real-world geographic and development challenges.

Course Outcomes:

After successful completion of the course, the students will be able to:

CO.	Course Outcome	Blooms Taxonomy Level
CO1	**Define** the importance of field techniques in geographic research and relate practical skills to employability goals of SDG 4.4.	BT1
CO2	**Explain** the design of field surveys, questionnaires, and sampling methods as tools of innovation in applied geography (SDG 9.5).	BT2
CO3	**Apply** field data collection and GPS/geospatial tools in conducting geographic surveys, linking skill development to SDG 4.4 and SDG 9.5.	BT3
CO4	**Analyse** field data and prepare geographic reports using systematic methods, demonstrating the application of employable research skills.	BT4

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Geographic Research: Definition and types, Research methodology in geography; Defining a research problem; Statement of the problem; Objectives, and hypothesis/ research questions, Database and methodology, significance, review of research works and bibliography and references, Research design: Meaning, need and features of a good design	10
Unit 2	Research ethics: Plagiarism- classification and prevention; Intellectual property rights; Research report: Structural components and presentation.	14
Unit 3	Field Tools and Techniques – Merits, Demerits and Selection of the Appropriate Technique; Observation (Participant / Non Participant), Questionnaires (Open/ Closed / Structured / Non-Structured) Interview with Special Focus on Focused Group Discussions; Space Survey (Transects and Quadrants, Constructing a Sketch), Surveying: Plane and Geodetic Surveying; Concept of ground surveying	14

Unit 4	Field Work in Geographical Studies – Role, Value, Data and Ethics of Field Work; Defining the Field and Identifying the Case Study – Rural / Urban / Physical / Human / Environmental; Planning and Conducting Field Survey based on Geographical Studies; Sampling Techniques and Survey Design; Collection of Primary Data through Household, Community and Environmental Surveys; Designing the Field Report – Aims and Objectives, Methodology, Analysis, Interpretation and Writing the Report. *Commencement of Field work will be in the last week of October.	10
	.	**48**

Text Books:

1. Creswell J., 1994: ***Research Design: Qualitative and Quantitative Approaches,*** Sage Publications.
2. Steers, J. A., 1965: ***An Introduction to the Study of Map Projection***, University of London, London

Reference Books:

1. Robinson A., 1998: *"Thinking Straight and Writing That Way", in Writing Empirical Research Reports: A Basic Guide for Students of the Social and Behavioural Sciences*, eds.by F. Pryczak and R. Bruce Pryczak, Publishing: Los Angeles. Special Issue on "Doing Fieldwork" The Geographical Review 91:1-2 (2001).
2. Stoddard R. H., 1982: *Field Techniques and Research Methods in Geography*, Kendall/Hunt.

Wolcott, H. 1995. *The Art of Fieldwork*. Alta Mira Press, Walnut Creek, CA

Minor	GEOGRAPHY OF NORTHEAST INDIA L-T-P-C: 4-0-0-4 Credit Units: 4 (T) Scheme of Evaluation:	Subject Code: GEO164N701

Course Objectives: This minor course introduces students to the physical, cultural, and economic geography of North East India, with a focus on the region's rich natural heritage. Aligned with **SDG 15: Life on Land** (sub-theme: 15.1 Forest conservation), the course fosters an appreciation for the region's forested landscapes, river systems, and indigenous land-use traditions, while examining the challenges of conserving terrestrial biodiversity amidst pressures of population growth and development.

Course Outcomes:

By the end of this course the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Define** the physical and human geographic features of North East India with reference to the significance of forest ecosystems (SDG 15.1).	BT1
CO2	**Explain** the patterns of land use, forest cover, and conservation practices in North East India aligned with SDG 15.1 goals.	BT2
CO3	**Identify** socio-economic and environmental challenges threatening forest conservation in the region.	BT3
CO4	**Analyse** regional development strategies in North East India for their compatibility with land conservation and sustainable ecosystem management under SDG 15.	BT4
CO5	**Examine** the basic idea of position of Northeast India and Assam in Indian context.	BT5

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	North East India: location and strategic significance; Physical characteristics and their relation to development: Relief, drainage, climate, soil and vegetation and bio-diversity, North-East India's relation to its neighbors.	12
Unit 2	Population and Development: Population growth, composition and distribution, migration, population characteristics, social structure: race, caste, religious and linguistic composition.	12
Unit 3	Natural resources, their utilization and development: Coal, petroleum, natural gas, water and forests in North East India; Agriculture and Development: Agricultural modernization and strategies for future development; constraints of Industrial development; problems and	12

	prospects of tourism.	
Unit 4	Locational significance of Assam; its physical characteristics and their relation to development: Relief, drainage, climate, soil and vegetation, Population growth, composition and distribution in North-East India and Assam; Characteristics and status of agriculture, industry and tourism in Assam	12
	Total	**48**

Text Books:

5. Dutta Ray, B., et. al (eds), 2000: Population, Poverty and Environment in North East India,
6. Concept Publishing Co., New Delhi.
7. Taher M. and Ahmed, P., 2000: Geography of North East India, Mani-Manik Prakash, Guwahati.
8. Bhagabati, A. K. et al, 2001: Geography of Assam, Rajesh Publications, New Delhi.

Reference Books:

2. Barua, P. C., 1990: Development Planning of North East India, Mittal Publications, New Delhi.

North East India Geographical Society: North Eastern Geographer, Department of Geography, Gauhati University.

Semester *8*

Major	RESEARCH METHODOLOGY L-T-P-C: 4-0-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEO162M801

Course Objectives: This course introduces students to the philosophical foundations, methods, and tools of geographic research. Aligned with **SDG 4: Quality Education** (sub-themes: 4.4 Skills for employment, 4.7 Education for sustainable development), the course cultivates systematic research thinking, technical competency in data collection and analysis, and the application of research methodologies to address contemporary geographic and developmental problems, thereby preparing graduates for research careers and lifelong learning.

Course Outcomes:

After the completion of course, the students will have ability to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	**Define** the philosophy, scope, and types of geographic research and relate them to skills for employability in research contexts (SDG 4.4).	**BT 1**
CO2.	**Explain** research design, hypothesis formulation, and methodological approaches in alignment with education for sustainable development (SDG 4.7).	**BT 2**
CO3.	**Apply** appropriate data collection instruments (questionnaires, field surveys, geospatial tools) to conduct geographic research.	**BT 3**
CO4.	**Analyse** geographic data using qualitative and quantitative techniques, demonstrating employment-ready research skills (SDG 4.4).	**BT 4**
CO5.	**Interpret** the various types of data along with **critical evaluation**.	**BT 5**
CO6.	**Design and develop** a scientific research report	**BT 6**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Research: Definition, types, significance and important features; Research methodology in geography; Defining a research problem; Statement of the problem; Objectives, and hypothesis/ research questions, Database and methodology, significance, review of research works and bibliography and references.	12
Unit 2	Research design: Meaning, need and features of a good design, Inductive and deductive approaches in geographic research, Hypothesis, theories and models in Geography; concept of model building and hypothesis testing.	12

Unit 3	Field techniques in Geography: Types, role and significance; Questionnaire design (Open, Closed, Structured, Non-structured), data collection, Post field processes: construction of data matrix, data processing and analysis; Introduction to qualitative techniques in Geography-observation, focus group discussion, case study, narratives, content analysis, oral history.	12
Unit 4	Sources of geographic data (Conventional and Geospatial technology based), their representation, interpretation and analysis; Research ethics: Plagiarism-classification and prevention; Intellectual property rights; Research report: Structural components and presentation.	12
	Total	**48**

Note: Computer / calculator based compulsory home assignments may be given for various units. Scientific calculator may be permitted in the examination hall for this paper.

Text Books:

1. Harvey, D, 1969: Explanation in Geography, Scientific Publisher, Jodhpur.
2. Lenon, B., Cleves, P. 2015. Geography Fieldwork and Skills, Harper-Collins.
3. Montello , D.R, Sutton, P. 2012. An Introduction to Scientific Research Methods in Geography and Environmental Studies, 2nd ed, Sage.
4. Murthy , K.L.N. 2004. Research Methodology in Geography: A Text Book, Concept Publishing Co.

Reference Books:

1. Evans, M., (1988): "Participant Observation: The Researcher as Research Tool" in Qualitative Methods in Human Geography, eds. J. Eyles and D. Smith, Polity.
2. Special Issue on "Doing Fieldwork" The Geographical Review 91:1-2 (2001).
3. Stoddard, R. H., (1982): Field Techniques and Research Methods in Geography, Kendall/Hunt.
4. Wolcott, H., (1995): The Art of Fieldwork, Alta Mira Press, Walnut Creek, CA.
5. Northey, N., Draper, D., Knight, D.B. 2015. Making Sense in Geography and Environmental Sciences:A Student's Guide to Research and Writing, 6th ed, Oxford University Press.
6. Parsons, T., Knight, P.G. 2015. How To Do Your Dissertation in Geography and Related Disciplines,3rd ed, Routledge.

MAJOR	RESEARCH I Credit Units: 4 Scheme of Evaluation: (P)	Subject Code: GEO162M822

Learning Objectives: This dissertation/research project course enables students to independently design and execute original geographic research, integrating theoretical knowledge with empirical fieldwork or data analysis. Grounded in **SDG 4: Quality Education** (sub-theme: 4.7 Education for sustainable development) and **SDG 17: Partnerships for the Goals** (sub-themes: 17.6 Knowledge sharing, 17.17 Partnerships), the course encourages research that contributes to sustainable development knowledge, promotes collaborative academic inquiry, and builds capacities for meaningful knowledge exchange and multi-stakeholder partnerships in geographic research.

Learning Outcomes:

After the completion of course, the students will have ability to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	**Define** a researchable problem in geography and frame it within a sustainable development context consistent with SDG 4.7.	**BT 1**
CO2.	**Design** a research methodology with appropriate data collection and analysis strategies, reflecting education for sustainable development principles (SDG 4.7).	**BT 2**
CO3.	**Conduct** original geographic research, applying interdisciplinary approaches and building knowledge-sharing capacities (SDG 17.6).	**BT 3**
CO4.	**Analyse** and interpret research findings through the lens of sustainability, contributing to the evidence base for SDG-aligned interventions.	**BT 3**
CO5.	**Appraise** and disseminate research outcomes effectively, fostering academic partnerships and collaborative learning consistent with SDG 17.17.	**BT 5**
CO6.	**Design and develop** a detail project report	**BT 6**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Identification of research problem / topic on any one of the following aspects (preferably of local area / Gram Panchayat / Revenue circle / District / State) during the 3rd semester while working with a minor project by the individual student:- Any kind of geographical studies on socio-economic / cultural issues / demographic problems and characteristics a. Agriculture, Industry, Mining related issues b. Environmental problems of the locality including disasters and hazards c. Natural resources assessments, planning and management	12

	d. Ecological crisis, Climate change and consequences e. Rural / Urban Ecosystems; f. Terrain / basin / watershed characterization and evaluation including integrated development studies *(This list is indicative only, the student may consult the assigned teacher as project supervisor / guide. Project supervisor / guide to each student will be allocated)*. The topic selection / modification may be done just before the 3rd Semester End Examination so that the data collection can be done during semester break. A fresh project proposal / modified project proposal of minor project done in 3rd semester is to be submitted by each student (within 2nd week of the commencement of 4th semester classes) by mentioning the following:- a. Project title b. Introduction to the problem c. Aims / objectives d. Research questions e. Database and Methodology f. Study of relevant literature g. Organization of study *Marks for internal evaluation = 14*	
Unit 2	Project proposal presentation by each student using PowerPoint during 3rd week of the commencement of the course of 4th semester. *Marks for internal evaluation = 14*	12
Unit 3	Reporting of data collection, tabulation, processing, mapping/charting and analysis by each student using PowerPoint during 5th week of the commencement of the course of 4th semester. *Marks for internal evaluation = 14*	12
Unit 4	Preparation of project report in prescribed format during 6th - 8th week of the commencement of course of 4th semester. Submission of the report after a week of the announcement of routine for 4th End Semester Examination. Final project presentation by each student using PowerPoint during on the scheduled date of viva-voce examination of this paper. *Marks for external evaluation = Viva-voce 8 + Presentation 20 = 28*	12
	Total	**48**

Note: Submission of project report in prescribed format and on specified date is mandatory. Equal weightages of marks for each stage of the work (upto 3rd stage) for internal evaluation of the project by the supervisor (60% of end semester examination). 40% of the total marks of end semester examination is for viva-voce and final presentation to be evaluated by an external examiner.

Text Books:

As per the list of given in syllabus based on topic selected

Reference Books:

As per the list of given in syllabus based on topic selected

Minor	ENVIRONMENT AND SUSTAINABLE DEVELOPMENT			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEO162N801

Course Objectives: This course provides students with a comprehensive understanding of the relationship between human activities, the environment, and the principles of sustainable development. Aligned with **SDG 12: Responsible Consumption and Production** (sub-themes: 12.2 Resource efficiency, 12.8 Sustainability awareness), **SDG 13: Climate Action** (sub-theme: 13.2 Mitigation and adaptation), and **SDG 15: Life on Land** (sub-theme: 15.1 Forest conservation), the course equips students to critically evaluate patterns of resource use, environmental degradation, and the policy frameworks necessary for achieving a sustainable balance between development and ecological conservation.

Course Outcomes:

By the end of this course the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	**Relate** to the basics of environmental change and sustainable development, connecting resource management principles to SDG 12.2 (Resource efficiency).	**BT 1**
CO2.	**Classify** different types of natural resources and assess their sustainable management in the context of responsible consumption (SDG 12.2).	**BT 2**
CO3.	**Develop** understanding about the impacts of unsustainable resource use on climate, land, and biodiversity, aligned with SDG 13.2 and SDG 15.1.	**BT 3**
CO4.	**Inspect** issues of adaptation and mitigation, solid waste management, and environmental legislation, evaluating their effectiveness in meeting SDG 13 and SDG 15 targets.	**BT 4**
CO5.	**Explain** the role of sustainability awareness (SDG 12.8) in shaping environmental policies, international agreements, and community-level conservation actions.	**BT 5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Introduction to Environment: definition and concept, types, Environmental factors, The Global Environment and its segments and development, Natural resources: renewable and non-renewable, land resources, water resource, forest resource and energy resource	12
Unit 2	Man- environment relationship: Historical perspectives on man's interaction with environment; population growth and environment; Man and atmosphere: Man as a factor of climatic change; Global environmental problems: Types and extent of environmental problems, area-specific major environmental issues and problems.	12

Unit 3	Concept of sustainability, concept of sustainable development, history, : relation among environment, economy and society, Pillars and principles of Sustainable development, Environmental Pollution- causes and effects, Nuclear Hazard and Human Health, Solid Waste Management, Climate Change and Global warming, ozone layer depletion and its impacts	12
Unit 4	Environmental protection movements: Chipko Movement, Silent Valley, Narmada Bachhao Andoloan; Environmental Legislation Programme in India: Wildlife Protection Act, Water Act, Forest Act, Air Act, Environmental Protection Act, International Agreement-Earth Summit, UNFCC, Montreal and Kyoto Protocol, Environmental Communication and Awareness	12
	Total	**48**

Text Books:

1. Goudie, A., (1984): The Nature of Environment, Basil Blackwell, London.
2. Singh, S., 1991: Environmental Geography, Prayag Pustak Bhawan, Allahabad

Reference Books:

1. Park, C., (1997): The Environment, Routledge, London.
2. Pickering, K. T. & L. A. Owne, (1994): An Introduction to Global Environmental Issues, Routledge, London.
3. .
4. Strahler, A. N. and A. H. Strahler, 1976: Geography and Man's Environment, John Willey, New York.

MAJOR	GLOBAL CLIMATE CHANGE L-T-P-C: 4-0-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEO162M804

Course Objectives: This course aims to provide students with a deep understanding of the science, impacts, and policy dimensions of global climate change. Anchored in **SDG 3: Good Health and Well-being** (sub-theme: 3.9 Environmental health), **SDG 13: Climate Action** (sub-themes: 13.1 Climate resilience, 13.2 Mitigation and adaptation), and **SDG 14: Life Below Water** (sub-theme: 14.2 Coastal ecosystems), the course enables students to critically assess the causes and consequences of climate change on human health, terrestrial and marine ecosystems, and to evaluate global and national strategies for mitigation and climate adaptation.

Course Outcomes:

After successful completion of the course, the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	**Relate** to the basics of climate change science and its documented impacts on environmental health as indicated by SDG 3.9.	**BT 1**
CO2.	**Classify** different types of climate vulnerability and explain their consequences for human health, food systems, and coastal ecosystems (SDG 3.9, 14.2).	**BT 2**
CO3.	**Develop** understanding about climate change impacts on agriculture, water resources, human health, and biodiversity, consistent with the goals of SDG 13.1 (Climate resilience).	**BT 3**
CO4.	**Inspect** the issues of adaptation and mitigation related to climate change, evaluating strategies for building resilience in coastal and terrestrial ecosystems (SDG 13.2, 14.2).	**BT 4**
CO5.	**Recommend** suitable and evidence-based measures for climate change mitigation aligned with national and international frameworks under SDG 13.	**BT 5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Causes of global warming and climate change - natural and anthropogenic (industrial and vehicular emissions of Green House Gases, Radiation patterns); Global precipitation and temperature distribution patterns; Extreme climatic events and processes; Urban heat island formation, concept of micro-climate in urban context El Nino and La Nina effects; Consequences of climate change in various geographical regions, concept of urban green space	12

Unit 2	Impact of climate change on agriculture and food security, human heath, migration and socio-economic issues; Geographical theories of climatic changes involving displacement of continents, change in the composition of atmosphere, solar radiation.	12
Unit 3	Global climatic assessment by Inter-governmental Panel on Climate Change (IPCC); Sources of climatic data and Statistical analysis of climatic data; Climate and biodiversity: Ecological succession of plants and animal life; Biogeographical shifts of bio-resources.	10
Unit 4	Global and national initiatives on adaptation and mitigation measures on climate change; National Action Plan on Climate Change; Role of Non-governmental Organizations, Local Institutions -Urban Local Bodies and Panchayats	14
	Total	**48**

Text Books:

1. Chritchfield, H. J., 1992: General Climatology, Prentice-Hall of India Pvt. Ltd, New Delhi
2. Lal, D. S., 1997: Climatology, Sharada Pustak Bhawn, Allahabad-02
3. Lal, D. S., 2016: Climatology and Oceanography, Sharada Pustak Bhawn, Allahabad-02
4. Strahler, Alan., 2018: Introducing Physical Geography, 6th Edition, Wiley India Pvt Ltd. New Delhi

Reference Books:

1. IPCC, 2014: Climate Change 2014: Impacts, Adaptation, and Vulnerability. Part A: Global and Sectoral Aspects. Contribution of Working Group II to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA.
2. IPCC, 2007: Climate Change 2007: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change.
3. OECD, 2008: Climate Change Mitigation: "What do we do?" (Organisation and Economic Co-operation and Development).
4. UNEP, 2007: Global Environment Outlook: GEO4: Environment for Development, United Nations Environment Programme.
5. Sen, Roy, S., and Singh, R.B., (2002): Climate Variability, Extreme Events and Agricultural Productivity in Mountain Regions, Oxford & IBH Pub., New Delhi.
6. Singh, R.B., Mal, Suraj, and Huggel, Christian (2018): Climate Change, Extreme Events and Disaster Risk Reduction, Springer, Switzerland.

MAJOR	ENVIRONMENT AND SUSTAINABLE DEVELOPMENT			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEO162M805

Course Objectives: This course provides students with a comprehensive understanding of the relationship between human activities, the environment, and the principles of sustainable development. Aligned with **SDG 12: Responsible Consumption and Production** (sub-themes: 12.2 Resource efficiency, 12.8 Sustainability awareness), **SDG 13: Climate Action** (sub-theme: 13.2 Mitigation and adaptation), and **SDG 15: Life on Land** (sub-theme: 15.1 Forest conservation), the course equips students to critically evaluate patterns of resource use, environmental degradation, and the policy frameworks necessary for achieving a sustainable balance between development and ecological conservation.

Course Outcomes:

By the end of this course the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	**Relate** to the basics of environmental change and sustainable development, connecting resource management principles to SDG 12.2 (Resource efficiency).	**BT 1**
CO2.	**Classify** different types of natural resources and assess their sustainable management in the context of responsible consumption (SDG 12.2).	**BT 2**
CO3.	**Develop** understanding about the impacts of unsustainable resource use on climate, land, and biodiversity, aligned with SDG 13.2 and SDG 15.1.	**BT 3**
CO4.	**Inspect** issues of adaptation and mitigation, solid waste management, and environmental legislation, evaluating their effectiveness in meeting SDG 13 and SDG 15 targets.	**BT 4**
CO5.	**Explain** the role of sustainability awareness (SDG 12.8) in shaping environmental policies, international agreements, and community-level conservation actions.	**BT 5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Introduction to Environment: definition and concept, types, Environmental factors, The Global Environment and its segments and development, Natural resources: renewable and non-renewable, land resources, water resource, forest resource and energy resource	12
Unit 2	Man- environment relationship: Historical perspectives on man's interaction with environment; population growth and environment; Man and atmosphere: Man as a factor of climatic change; Global environmental problems: Types and extent of environmental problems, area-specific major environmental issues and problems.	12

Unit 3	Concept of sustainability, concept of sustainable development, history, : relation among environment, economy and society, Pillars and principles of Sustainable development, Environmental Pollution- causes and effects, Nuclear Hazard and Human Health, Solid Waste Management, Climate Change and Global warming, ozone layer depletion and its impacts	12
Unit 4	Environmental protection movements: Chipko Movement, Silent Valley, Narmada Bachhao Andoloan; Environmental Legislation Programme in India: Wildlife Protection Act, Water Act, Forest Act, Air Act, Environmental Protection Act, International Agreement-Earth Summit, UNFCC, Montreal and Kyoto Protocol, Environmental Communication and Awareness	12
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Text Books:

3. Goudie, A., (1984): The Nature of Environment, Basil Blackwell, London.
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Reference Books:

1. Park, C., (1997): The Environment, Routledge, London.
5. Pickering, K. T. & L. A. Owne, (1994): An Introduction to Global Environmental Issues, Routledge, London.
6. .
7. Strahler, A. N. and A. H. Strahler, 1976: Geography and Man's Environment, John Willey, New York.

MAJOR	GEOGRAPHY OF HEALTH L-T-P-C: 3-1-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEO162M806

Course Objectives: This course introduces students to the geographic dimensions of human health and disease, examining how spatial, environmental, and socio-economic factors shape health outcomes across regions. The course is linked to **SDG 3: Good Health and Well-being** (sub-themes: 3.3 Disease prevention, 3.8 Health systems, 3.9 Environmental health), enabling students to understand the geographic distribution of diseases, the role of environmental conditions in health, and the spatial inequalities in access to healthcare systems, contributing towards a geographically informed approach to achieving universal health goals.

Course Outcomes:

By the end of this course the students will be able to:		
CO	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Recall** different disease classifications and geographic approaches to health, relating them to disease prevention goals of SDG 3.3.	BT1
CO2	**Outline** the concepts of human health and healthcare from a geographic perspective, linking spatial disparities to health systems strengthening under SDG 3.8.	BT2
CO3	**Develop** knowledge about the environmental and ecological factors influencing human health across varying geographic settings, consistent with SDG 3.9 (Environmental health).	BT3
CO4	**Analyse** the impact of environmental degradation and pollution on the occurrence and spatial distribution of diseases, applying the framework of SDG 3.9.	BT4

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Geography of Health: Definition and significance; approaches of study: ecological, social, and spatial; dualism between medical geography and geography of health. Classification of diseases: genetic, zoonotic, communicable, non-communicable, occupational, deficiency diseases and malnutrition.	10
Unit 2	Disease ecology: ecology and human health; geographical factors affecting human health; factors influencing disease transmission (pathological, physical, environmental, social, cultural, and economic); Diffusion of diseases and their causes in varied biotic, physical, and cultural environments.	14

Unit 3	Disease occurrence: emergence, re-emergence, and persistence; modes of transmission of major diseases (Malaria, Japanese encephalitis, tuberculosis, hepatitis, AIDS, and COVID-19) and their broad global distribution. Healthcare systems: Meaning and components; Universal government-funded health system; Role of WHO and UNICEF in global health care; SDG3 for good health and Well-being.	14
Unit 4	Environment, human habit, and health: Basic concept and ideas relating to food habit and health, occupation and health, environmental degradation and health, lifestyle, and human health.	10
	Total	**48**

Text Book:

1. Akhtar Rais (Ed.), 1990: Environment and Health Themes in Medical Geography, Ashish Publishing House, New Delhi
2. Anthamatten P, (2011), Introduction to the Geography of Health, Rawat Publications, Jaipur

References:

1. Cliff, A.D. and Peter,H., 1988 : Atlas of Disease Distributions, Blackwell Publishers, Oxford
2. Hardham T. and Tannav M.,(eds): Urban Health in Developing Countries; Progress, Projects, Earthgoan, London
3. Murray C. and A. Lopez, 1996: The Global Burden of Disease, Harvard University Press.
4. National Health Portal India https://www.nhp.gov.in/healthprogramme/national-health programmes
5. Shaw, M., Dorling, D. and Mitchell, R, (2002) Health, Place and Society, Pearson, London Brown, T., McLafferty, S., Moon, G. (2010): A Companion to Health and Medical Geography, Wiley Blackwell, UK
6. Mishra, R.P. (1970): Medical Geography of India, National Book Trust of India

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